RESEARCH ANALYSIS AND APPLICATION MAPPING

Holistic Health Optimization Platform

A research-grounded reference mapping six primary studies and twenty-four verified supporting sources onto the platform's four components: Prakriti and Vikriti assessment, smartwatch integration, and an evidence-labelled Ayurveda recommendation engine.

Research & Development Reference

Prepared for the platform project team and mentor review - paper-by-paper analysis, questionnaire scoring models, wearable validation evidence, knowledge-base design and a prioritised roadmap.

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Table of Contents

1. Project Overview 1

1.1 The Four Application Components 1

1.2 Scope, Intent and Non-Goals 2

2. Research Methodology and Paper Selection 2

2.1 How Sources Were Selected and Accessed 2

2.2 Selection Criteria and Evidence Typing 3

3. Executive Summary 3

4. Paper-by-Paper Analysis 5

4.1 Paper 1 - Gupta et al. (2025): Standardizing Prakriti Evaluation (Scoping Review) 5

A. Paper information 5

B. Research objective 5

C. Methodology 5

D. Important findings 6

E. Limitations 6

F. Direct application to the project 7

4.2 Paper 2 - Tripathi et al. (2011): Cardiovascular Responses and the 62-Item Self-Assessment Questionnaire 7

A. Paper information 7

B. Research objective 7

C. Methodology 8

D. Important findings 8

E. Limitations 9

F. Direct application to the project 9

4.3 Paper 3 - Bhalerao & Patwardhan (2016): Good Reporting Practices for Prakriti Research 9

A. Paper information 9

B. Research objective . . . 10
C. Methodology and content . . . 10
D. Important findings . . . 10
E. Limitations . . . 10
F. Direct application to the project . . . 11
4.4 Paper 4 - Mills et al. (2019): Vikriti Questionnaire and Psychological States . . . 11
A. Paper information . . . 11
B. Research objective . . . 11
C. Methodology . . . 12
D. Important findings . . . 12
E. Limitations . . . 12
F. Direct application to the project . . . 13
4.5 Paper 5 - Anagha & Deole (2021): Dinacharya, the Daily Regimen (CSNE Book Chapter) . . . 13
A. Paper information . . . 13
B. Research objective . . . 13
C. Methodology and content structure . . . 14
D. Important findings . . . 14
E. Limitations . . . 15
F. Direct application to the project . . . 16
4.6 Paper 6 - Dhiman et al. (2026): Ritucharya Critical Narrative Review (Abstract-Level Analysis) . . . 16
A. Paper information . . . 16
B-E. What the verified abstract establishes . . . 17
F. Direct application to the project . . . 17
4.7 Supplementary Sources Analysed for This Project . . . 17
5. Prakriti Research Findings . . . 19
5.1 What Prakriti Is, Operationally . . . 19
5.2 Assessment Approaches and Their Evidence . . . 19

5.3 The Scoring Model the Platform Should Implement . . . 19
5.4 Psychometric Targets for the Platform's Own Validation . . . 20
6. Vikriti Research Findings . . . 20
6.1 What Vikriti Is, Operationally . . . 20
6.2 The 60-Item Questionnaire: Structure, Scoring, Domains, Limits . . . 20
6.3 What the Platform May and May Not Claim . . . 21
6.4 Assessment Frequency and State Confounders . . . 21
7. Prakriti vs Vikriti Comparison . . . 21
8. Smartwatch Integration Research . . . 22
8.1 Variable-by-Variable Evidence Table . . . 22
8.2 Directly Measured vs Estimated - A Non-Negotiable Distinction . . . 23
8.3 Platform Access: Android Health Connect . . . 23
8.4 Platform Access: Apple HealthKit . . . 23
8.5 Synchronization, Missing Data, Timestamps and Consent . . . 24
8.6 What the App Must Not Do . . . 24
9. Ayurveda Knowledge Extraction . . . 24
9.1 Dinacharya (dinacaryā) - Daily Routine . . . 25
9.2 Ritucharya (ṛtucaryā) - Seasonal Routine . . . 25
9.3 Ahara (āhāra) - Food and Eating Habits . . . 26
9.4 Vyayama (vyāyāma) - Exercise . . . 26
9.5 Nidra (nidrā) - Sleep . . . 26
9.6 Sadvritta (sadvṛtta) - Conduct and Mental Wellness . . . 27
9.7 Cross-Cutting: Dinacharya's Constitution-Conditional Pattern . . . 27
10. Sanskrit Terminology and Recommendation Database . . . 27
10.1 Recommendation Record Schema (Conceptual) . . . 28

11. Research Gap and Novelty Analysis . . . 28

11.1 What Already Exists . . . 29

11.2 Comparison of Existing Systems . . . 29

11.3 Limitations That Remain and What Has Not Been Studied . . . 30

11.4 Realistic Novelty Contributions for This Platform . . . 30

12. Proposed System Architecture . . . 31

12.1 Component Architecture . . . 31

12.2 Module Responsibilities . . . 32

12.3 Data Flow and Trust Boundaries . . . 32

12.4 Proposed Data Schemas (JSON) . . . 32

13. Research-to-Application Implementation Mapping . . . 33

14. Sample Recommendation Rules . . . 35

14.1 Engine Design Principles . . . 35

14.2 Rule Record Format . . . 35

14.3 Twelve Worked Example Rules . . . 35

14.4 Handling Missing and Conflicting Inputs . . . 36

15. Limitations and Validation Requirements . . . 37

15.1 Evidence Limitations the Team Must Own . . . 37

15.2 Validation Requirements Before Each Release Stage . . . 37

15.3 What to Tell the Mentor (Evidence and Limits Summary) . . . 37

16. Prioritized Development Roadmap . . . 38

16.1 Phase Plan . . . 38

16.2 Deliberate Non-Goals for v1 . . . 38

17. Bibliography . . . 38

Core papers (all read in full unless noted) . . . 38

Supporting peer-reviewed sources . . . 39

Platform documentation and classical sources 40

1. Project Overview

The Holistic Health Optimization Platform is an educational wellness application that combines four research-grounded components: a Prakriti (baseline constitution) assessment engine, a Vikriti (current-state imbalance) assessment engine, a smartwatch integration module, and an Ayurveda-informed recommendation engine covering daily routine (dinacaryā), food and eating habits (āhāra), exercise (vyāyāma), seasonal routine (ṛtucaryā), sleep (nidrā), mental wellness and conduct (sadvṛtta). The platform is explicitly designed as an educational wellness tool, not a medical diagnostic or treatment system, and this boundary shapes every design decision analysed in this report. This document maps specific, verifiable research findings onto each of the four components so that the development team can trace every feature back to a source.

The four components operate at different time scales, and the distinction matters for software design. Prakriti is conceptualised in Ayurveda as the relative proportion of the three doṣas (Vata, Pitta, Kapha) established at conception and believed to remain stable across the lifespan [1, 2]. Vikriti represents the current, fluctuating state of the doṣas, which responds to diet, season, stress and behaviour [1, 4]. Wearable signals (heart rate, heart-rate variability, sleep, activity) provide continuously sampled physiological context at minutes-to-days resolution [10, 14, 19]. Seasonal context operates at weeks-to-months resolution [6, 21]. The recommendation engine must therefore treat Prakriti as the stable personalisation anchor, Vikriti as the adaptive signal, wearable trends as objective context, and season as a modifier - without collapsing these four layers into a single score, a step for which no reviewed study provides scientific justification.

1.1 The Four Application Components

Component A - Prakriti assessment. The platform currently plans a 62-question Prakriti questionnaire producing Vata, Pitta and Kapha percentage scores and a dominant or dual-doṣa classification. Chapter 4.2 and Chapter 5 document the direct scientific lineage of this instrument: the self-assessment questionnaire used by Tripathi, Patwardhan and Singh [2] is recorded by the 2025 scoping review of Gupta et al. [1] as a 62-item, Charaka-based self-assessment tool with criterion-validity evidence, and it is one of the better-cited instruments in the field. The questionnaire is anchored in the guṇa (attribute) logic of the Caraka-saṃhitā, Vimāna-sthāna 8.95 [3, 26].

Component B - Vikriti assessment. The platform needs a separate, research-supported method for assessing the current state. Chapter 4.4 and Chapter 6 analyse the 60-item vikṛti questionnaire used by Mills et al. [4], which is currently the best-documented vikṛti instrument in the peer-reviewed literature, together with the psychological-state correlations that support its use as a wellness-monitoring input rather than a diagnostic instrument. Chapter 7 sets out the Prakriti-Vikriti distinction in the operational form the development team needs.

Component C - Smartwatch integration. Per the mentor's instruction, the platform reads measurements from a connected smartwatch rather than collecting them manually. Chapter 8 evaluates each candidate variable (heart rate, resting heart rate, HRV, steps, active minutes, sleep duration, sleep stages, SpO_2, skin temperature, respiratory rate) against validation studies, and evaluates the Android Health Connect and Apple HealthKit access layers, their permission models and their known limitations [24, 25]. The chapter distinguishes directly measured values from device estimates throughout, because this distinction determines which variables the recommendation engine may trust.

Component D - Ayurveda recommendation engine. Chapters 9-11 build a structured knowledge database from the reviewed literature and classical sources, in which every recommendation carries its Sanskrit name, transliteration, explanation, classical source, research reference, applicable context, an implementable rule sketch, and precautions. Chapter 14 turns this database into an explainable, rule-based engine design with twelve worked example rules, and Chapter 13 maps every finding onto the planned React + Vite, FastAPI, JSON-rules, Pytest technology stack.

1.2 Scope, Intent and Non-Goals

This report is a research-to-engineering bridge document, not a general literature review. Its objective is to answer, for each module of the platform, exactly which questionnaire, scoring method, variable, rule or methodology can be adopted from the reviewed sources, what evidence supports it, and what must be avoided or left unvalidated. Three boundaries are maintained throughout. First, the platform makes no medical claims: no reviewed paper supports the use of a consumer smartwatch as a direct measurement of an Ayurvedic doṣa, and this report does not construct such a claim. Second, traditional recommendations are labelled as traditional unless reviewed research actually supports them; correlations are not treated as causation. Third, where a source could not be accessed in full - specifically Dhiman et al. (2026) [6], whose publisher sits behind bot-protection - the analysis relies only on the publisher-verified abstract and metadata, and this limitation is stated explicitly rather than papered over.

The report also deliberately avoids pretending that software exists before it does. Module names and file paths proposed in Chapters 12-14 are suggestions for the planned React + Vite / FastAPI / JSON-rules / Pytest stack, not descriptions of an existing repository, because no repository listing was supplied with the request. Where the report proposes numerical parameters (for example, 7-day rolling windows for wearable trends), they are labelled as design proposals to be calibrated, not research-derived constants.

2. Research Methodology and Paper Selection

2.1 How Sources Were Selected and Accessed

The analysis began with the six references supplied by the project team and extended outward to peer-reviewed supporting literature. Each source was located through its DOI or PubMed record; full texts were retrieved and read where openly accessible. For every claim carried into this report, the citation points to a source the authors of this report actually accessed, and where a finding comes from a specific table, figure or section of a paper, that location is named. Papers whose full text could not be opened are identified as such, and only publisher-verified metadata and abstracts are used for them.

The six primary papers were accessed as follows. Gupta et al. (2025) [1] was read in full from PubMed Central (PMC12269270), including its seven tables and its discussion and conclusions. Tripathi et al. (2011) [2] was read in full from PubMed Central (PMC2952295), including the questionnaire scoring model, validation statistics, all test protocols, results tables and discussion. Bhalerao and Patwardhan (2016) [3] was read in full from PubMed Central (PMC4910291). Mills et al. (2019) [4] was read from PubMed Central (PMC6822152), including methodology, results and stated limitations. Anagha and Deole (2021) [5] is a peer-reviewed book chapter of the Charak Samhita New Edition (CSNE) published by the Charak Samhita Research, Training and Skill Development Centre; it was read in full from the publisher's open site (carakasamhitaonline.com), and its classical verse citations were extracted section by section. Dhiman et al. (2026) [6] could not be opened in full: the publisher site (journalajmah.com) blocks automated access, and the full text is not deposited in any open archive this project could locate. Only its Crossref-verified metadata and the complete publisher abstract were used, and every statement attributed to it in Chapter 4.6 comes from that abstract.

Supplementary literature was retrieved through PubMed and publisher APIs, prioritising systematic reviews, scoping reviews, position papers and instrument validation studies. All bibliographic details (journal, year, volume, pages, DOI) were verified against PubMed/Crossref records rather than copied from secondary listings. Developer documentation for Android Health Connect and Apple HealthKit was read from the official developer portals in October 2026 [24, 25]. Classical Ayurvedic sources are cited at the level of chapter and verse exactly as given by the accessed modern

sources [3, 5]; where two classical texts describe the same attribute differently, the disagreement is reported rather than silently resolved, following the reporting practice recommended by Bhalerao and Patwardhan themselves [3].

2.2 Selection Criteria and Evidence Typing

Sources were retained when they met at least one of three criteria: (a) they describe or validate an instrument the platform intends to implement (Prakriti or Vikriti assessment); (b) they report validation evidence for a wearable measurement the platform intends to collect; or (c) they provide the classical-textual or review-level grounding for a recommendation category the engine must output. Predatory or non-peer-reviewed sources were excluded, following the screening logic used by Gupta et al. [1]. Because the platform is an educational wellness tool, each imported finding is tagged in this report with one of four evidence types, and the recommendation engine in Chapter 14 carries these tags as data:

- **Demonstrated result:** a quantitative finding reported in a peer-reviewed paper (e.g., the inter-rater reliability range of 0.20-0.40 reported for Ayurveda physicians in [1], Section 4).
- **Author interpretation:** a conclusion the authors draw from their data without a direct statistical demonstration (e.g., Tripathi et al.'s hypothesis that a secondary doṣa can nullify the cardiovascular expression of the primary doṣa in dual types [2], Discussion).
- **Classical/traditional:** a recommendation grounded in classical texts, carried by a modern review (e.g., all dinacaryā components in [5]).
- **Design proposal:** an engineering suggestion of this report with no direct empirical source, always labelled (e.g., 7-day wearable trend windows).

This tagging is not cosmetic. Chapter 15 (Limitations and Validation Requirements) argues that the platform's scientific defensibility depends on keeping demonstrated results, interpretations and traditional claims visibly separate, both in the knowledge base and in the user-facing explanations generated by the recommendation engine.

3. Executive Summary

This section condenses the report for readers who need the decisions first and the evidence later. Nine findings dominate everything that follows, and each maps to a concrete module decision.

1. The 62-question Prakriti instrument has identifiable, citable scientific lineage - and known limits. The self-assessment questionnaire behind the platform's 62-question assessment derives from Patwardhan and Sharma's Charaka-based instrument, was modified and used by Tripathi et al. [2] with an explicit guṇa-proportional scoring model (maximum scores: Vata 960, Pitta 600, Kapha 1440), and is catalogued by Gupta et al. [1, Table 2] as a 62-item open-source tool with criterion validity (validation against a departmental questionnaire, n = 50; Vata r = 0.305, p = 0.015; Kapha r = 0.341, p = 0.031; Pitta r = 0.237, p = 0.097, not significant). It is therefore usable as the platform's baseline assessment, but it should be presented with its correlation-level validation, not as a fully psychometrically validated scale [1, 2].

2. AyuSoft and the Brief Prakriti Inventory are the strongest external benchmarks. The 2025 scoping review concludes that questionnaires remain the most validated Prakriti tools, names AyuSoft (85 items, inter-rater reliability assessed with 3,416 participants, proprietary) as the most rigorously validated, followed by the open-source Prototype Prakriti Analysis Tool (PPAT, 54 items) [1, Tables 2, 7; Section 5]. The 21-item Brief Prakriti Inventory (BPI) reports internal consistency of 0.75-0.87, test-retest ICCs of 0.83-0.90 and correlation of 0.78-0.84 with AyuSoft in a sample of 1,857 [8]. Both give the project concrete psychometric targets and, in BPI's case, an open comparability anchor.

3. Keep Prakriti and Vikriti as separate engines with separate instruments. Gupta et al. explicitly scope their review to Prakriti and exclude Vikriti as a distinct, fluctuating construct [1, Introduction], and Mills et al. treat vikṛti imbalance scores as state measures with demonstrated associations to psychological states: Vata imbalance with anxiety ($p \leq 0.05$) and rumination ($p \leq 0.01$), Pitta imbalance with poorer mood ($p \leq 0.01$) and stress, Kapha imbalance with stress ($p \leq 0.01$) [4, Results]. No reviewed source supports merging the two into one score.

4. The Vikriti instrument is documented but not validated as a diagnostic. The 60-item Chopra Center vikṛti questionnaire uses a three-point frequency scale (scores 5/3/1), higher scores indicating more imbalance, and its three dosha scales reduce to single factors explaining roughly 69-74% of variance [4, Methods, Results]. Its associations are exploratory, the sample was 101 relatively healthy adults, and Mills et al. themselves note the absence of comparison with traditional diagnostic methods [4, Limitations]. The platform may use this structure as its Vikriti input with clear framing.

5. For wearables, trust steps and heart rate, treat energy expenditure with suspicion, and treat sleep stages cautiously. Meta-analytic evidence shows Fitbit-class devices underestimate heart rate, steps and energy expenditure, with acceptable average accuracy for steps and heart rate but not energy expenditure [19]. In a 60-person validation cohort no device reached energy-expenditure error below 20%, while six devices achieved median heart-rate error below 5% during cycling, with error rising for darker skin tone, higher BMI and walking [10]. Sleep-wake summary metrics track well (epoch-by-epoch sleep sensitivity 0.93-0.96 across four devices at home [14]), but sleep-stage tracking remains the weak point, with macro-F1 scores ranging 0.26-0.69 across 11 consumer trackers against polysomnography [15].

6. Health Connect and HealthKit make the mentor's integration directive feasible, with defined gaps. Health Connect exposes typed records for heart rate, resting heart rate, RMSSD-based HRV, steps, sleep sessions with stages, oxygen saturation, skin temperature and respiratory rate, under declared permissions and a mandatory Play Console health-apps declaration, with background and >30-day historical reads requiring separate permissions [24]. HealthKit provides a user-permissioned central repository with read/share authorisation and multi-source merging [25]. Neither platform guarantees every device writes every record type; the app must degrade gracefully per variable.

7. No reviewed evidence maps smartwatch signals onto doṣas. This report found no peer-reviewed study establishing that any wearable variable directly measures Vata, Pitta or Kapha. The closest defensible linkage is indirect: dosha-grouped cardiovascular reactivity studies exist but are largely null - Tripathi et al. found basic cardiovascular responses did not vary significantly across dual-constitution groups except one post-exercise diastolic finding [2] - and therefore the engine must present wearable trends as wellness context, never as doṣa readouts.

8. The Ayurveda knowledge base can be built almost entirely from two open, citable sources. Anagha and Deole's peer-reviewed dinacaryā chapter [5] supplies the complete daily-regimen catalogue with classical verse citations (Charaka, Sushruta, Ashtanga Hridaya, Ashtanga Sangraha) and even constitution-specific itemisations (e.g., doṣa-specific tooth-stick tastes), while Dhiman et al.'s abstract [6] frames ṛtucaryā with the caution that its direct clinical evidence is sparse and constitution-based individualisation rests on preliminary data - a framing this report adopts verbatim as the platform's honesty posture for seasonal features.

9. The realistic novelty is disciplined integration, not a first-of-kind claim. The gap analysis (Chapter 11) finds Prakriti tools, Vikriti tools, wearable wellness apps and Ayurveda apps each exist separately, with no reviewed system joining them under explicit, explainable rules with per-recommendation evidence levels. A 2026 review of AI in Prakriti assessment [22] confirms standardisation and explainability as open needs. The platform can contribute: (i) an evidence-tagged, rule-based recommendation engine; (ii) strict Prakriti/Vikriti separation with dual-engine UX; (iii) honest wearable-context displays that never claim doṣa correspondence; (iv) a reproducible JSON rule schema; and (v) an evaluation protocol using validated instruments as comparators - all verifiable contributions that stop short of

unprovable priority claims.

4. Paper-by-Paper Analysis

Each of the six primary papers is analysed in a fixed format: paper information, research objective, methodology, findings (with source locations and evidence typing), limitations, and a direct-application table mapping every usable element onto a platform module. Evidence labels used in the tables: DR = demonstrated result, AI = author interpretation, CT = classical/traditional statement, DP = design proposal of this report. Supplementary sources are analysed in Section 4.7.

4.1 Paper 1 - Gupta et al. (2025): Standardizing Prakriti Evaluation (Scoping Review)

A. Paper information

- **Title:** Towards standardization of Prakriti Evaluation: A scoping review of modern assessment tools and their psychometric properties in Ayurvedic medicine.
- **Authors:** Ankit Gupta, Varsha Singh, Sushil Chandra, Rahul Garg (IIT Delhi; Rishihood University).
- **Journal/year:** Journal of Ayurveda and Integrative Medicine, 2025;16(4):101157. Published 2025 Jun 27.
- **Access:** DOI 10.1016/j.jaim.2025.101157; full text PMC12269270 (open access, CC BY-NC-ND). Read in full.
- **Research type:** Scoping review (Arksey and O'Malley five-stage framework; PRISMA flow diagram in Fig. 1).

B. Research objective

The authors ask two questions (Section 1, Stage 1): what questionnaires, devices, algorithmic and mathematical models exist for objectively assessing Prakriti, and what are the effectiveness and psychometric properties of each instrument? The motivation is that classical assessment (physical examination, pulse diagnosis, inference) produces subjective variation and bias among practitioners [1, Introduction], while no prior review systematically evaluated the available modern tools. Notably, the authors state that the review deliberately excludes Vikriti, "which represents the current state of doshas and can fluctuate" - language this report treats as direct textual support for keeping the platform's two assessments separate [1, Introduction, end].

C. Methodology

Studies published 2016-2022 assessing Prakriti via questionnaires, algorithms or devices were searched in Google Scholar (first ten result pages), PubMed and Scopus (899 records identified; 205 PubMed, 175 Scopus, 519 Google Scholar). After deduplication (425), title/abstract screening (125), and full-text screening with predatory-journal exclusion (Beall's list and UGC Care list; 39 studies excluded as predatory), 48 eligible studies remained; 24 were excluded because authors could not be contacted for their instruments, and 5 for reusing prior questionnaires, giving 19; backward citation search added 13, for a final 32 studies [1, Section 2, Fig. 1]. Screening was single-reviewer due to time constraints, a limitation the authors acknowledge [1, Section 2]. Tools were classified into questionnaires (16 studies), algorithms (5) and devices (11). Psychometric evaluation used internal consistency, test-retest and inter-rater reliability plus content, criterion and construct validity for questionnaires; validation methodology and standard metrics (accuracy, precision, recall, F-measure, sensitivity, specificity) for machine learning; and research reproducibility for devices [1, Section 2.1].

D. Important findings

Finding	What the evidence shows	Location / type
32 studies; questionnaires dominate	16 questionnaire studies, 5 algorithm-based, 11 device-based; questionnaires are the most commonly used and the most validated Prakriti tools.	Abstract; Tables 1-6; Section 3. DR
Few questionnaires are fully validated	Of 16 questionnaire studies: 3 validated, 6 with established reliability, 3 with both reliability and validity confirmed.	Abstract; Table 7. DR
AyuSoft is the best-validated tool	AyuSoft (Rotti et al. 2014): 85 items, Charaka + Sushruta + Ashtanga Hridaya base, inter-rater reliability, validation sample n = 3,416; proprietary licence. Conclusion names it the most rigorously validated, followed by open-source PPAT (54 items, IRR, content + construct validity, n = 26).	Tables 2, 7; Section 5 (Conclusion). DR
Tripathi et al. (2011) entry	Recorded as a 62-item self-assessment questionnaire, Charaka-based, scoring available, criterion validity, validation sample 50, open source.	Table 1 (SQ10), Table 2 (SQ10), Table 7. DR
Mysore Tridosha Scale entry	157-item psychological scale; internal consistency, split-half and test-retest reliability; content validity; validation samples 18 & 10 / 1,548 / 60; proprietary.	Table 1 (SQ11), Table 2, Table 7. DR
Other validated instruments	TNMC Prakriti 2004 (37 items; construct validity, >90% concordance with two physicians in 30 participants); Prasher et al. 2008 (73 items; IRR, n = 320); Suchitra et al. 2014 Child Prakriti Inventory (135 items; IC + SHR; CTV + CSV, n = 230); Edwards & Streiner 2022 (39 items; IC + TRR, n = 76 / 267, open).	Sections 3.1.1; Tables 2, 7. DR
Item count and stability framing	Questionnaire items range 20-157; only 8 of 16 questionnaires explicitly instruct respondents to answer for long-standing traits to capture Prakriti rather than current dosha state (vikriti).	Section 3.1.1. DR
Physician agreement is weak	Inter-rater reliability among Ayurveda physicians is typically 0.20-0.40, undermining "physician consensus" as ground truth.	Section 4 (Discussion). DR
No robust ground truth exists	The central obstacle to standardisation is absence of a reliable way to establish an individual's true Prakriti; proposals include a standardised ground-truth protocol, open labelled datasets, and correlating Prakriti with Western constructs (Big Five, somatotypes).	Sections 4, 4.1. AI
ML and devices immature	Machine-learning models show promise but suffer accuracy, replicability and ground-truth problems; of 11 device studies only one (SD11, wrist pulse acquisition system) provided validation data on request; none appropriately validated.	Abstract; Tables 4, 6; Section 5. DR

Table 4.1 Key findings of Gupta et al. [1] with source locations and evidence type.

E. Limitations

The authors acknowledge single-reviewer screening, the 2016-2022 publication window, and that Google Scholar coverage was capped at ten result pages [1, Section 2]. The review itself notes the field's deeper problems: no universally accepted Prakriti definition, weak construct validation in most questionnaires, unreported ground-truth procedures where multiple physicians were used, and datasets that are rarely open [1, Section 4]. For the platform, the practical consequence is that every questionnaire choice inherits field-level uncertainty; none of the instruments, including AyuSoft, resolves the ground-truth problem.

F. Direct application to the project

Finding from paper	Information extracted	App module	How to use it	Evidence strength / limitation
Tool landscape (Tables 1-7)	Complete inventory of 16 questionnaires with item counts, scoring availability, reliability/validity types, licences.	Prakriti assessment engine	Select the platform's instrument from this inventory; justify the 62-question choice against AyuSoft/PPAT/BPI; document the decision per Bhalerao's reporting checklist [3].	DR; scoping review, single-reviewer screening. Strong coverage, not a meta-analysis.
Tripathi 62-item entry (SQ10)	Confirms 62 items, Charaka basis, scoring pattern, criterion validity, open source.	Prakriti assessment engine	Use as external confirmation that the platform's 62-question instrument is citable; cite [1, Tables 2, 7] plus [2] together.	DR; criterion validity only (correlation with another questionnaire), no predictive validity.
AyuSoft validation record	85 items; IRR with n = 3,416; proprietary.	Prakriti engine; benchmarking plan	Use as the psychometric benchmark the platform should report against; do not copy the proprietary item set without licence.	DR; proprietary licence restricts reuse; IRR type not fully described in review.
Only 8/16 questionnaires anchor to stable traits	Wording must instruct users to answer for lifelong tendencies, not current states.	Prakriti questionnaire UX	Add explicit "over your adult life" instructions to every item; A/B-check response drift.	DR for the count; implementation is DP.
Physician IRR 0.20-0.40	Physician consensus is a weak gold standard.	Validation plan; user-facing claims	Avoid claiming physician-level equivalence; design a self-consistency + retest validation instead [3].	DR; range reported in review's discussion citing three primary studies.
Prakriti/Vikriti scoping statement	Review explicitly excludes Vikriti as a fluctuating, distinct construct.	Both engines	Quote this in project documentation to justify two separate instruments and databases.	DR (textual). Scope statement, not an empirical finding.

Table 4.2 Application mapping for Paper 1.

4.2 Paper 2 - Tripathi et al. (2011): Cardiovascular Responses and the 62-Item Self-Assessment Questionnaire

A. Paper information

- **Title:** The Basic Cardiovascular Responses to Postural Changes, Exercise, and Cold Pressor Test: Do They Vary in Accordance with the Dual Constitutional Types of Ayurveda?
- **Authors:** Piyush Kumar Tripathi, Kishor Patwardhan, Girish Singh (Dept. of Kriya Sharir and Biostatistics, Institute of Medical Sciences, Banaras Hindu University).
- **Journal/year:** Evidence-Based Complementary and Alternative Medicine, 2011;2011:251850 (accepted 2010, published online 2010 Aug 30).
- **Access:** DOI 10.1155/2011/251850; full text PMC2952295 (open access, CC BY). Read in full.
- **Research type:** Observational experimental physiology study with a questionnaire-development component.

B. Research objective

Prior Prakriti studies concentrated on extreme (single-doṣa) types, which the authors observe to be rare in practice; they therefore studied only dual-constitutional (dvandvaja) types and asked whether simple, non-invasive cardiovascular measures - heart rate and arterial blood pressure responses to posture change, isometric and isotonic exercise, and cold pressor test - vary by constitutional type. If they did, such measures might help verify Prakriti and indicate disease predisposition [2, Sections 1-2]. The study is equally valuable to this project for its questionnaire methodology, described with unusual completeness.

C. Methodology

Questionnaire. The instrument is a modified version of the "Self-assessment questionnaire for determining Prakriti" of Patwardhan and Sharma, built strictly from the Caraka-saṃhitā description (Vimāna-sthāna) so that each guṇa (attribute) and its observable features become yes/no questions. Charaka assigns 8 guṇas to Vata, 5 to Pitta and 12 to Kapha; each guṇa produces one or more observable features. To make dominance quantitatively comparable across doshas despite unequal guṇa counts, the authors allotted 120 score-points per guṇa, divided equally among that guṇa's features, giving maximum scores of 960 (Vata), 600 (Pitta) and 1,440 (Kapha). Dosha dominance (%) = (score obtained ÷ maximum for that dosha) × 100 - i.e., an absolute-basis scoring in the terminology of Bhalerao and Patwardhan [3, Scoring pattern]. Respondents self-score and report only percentages, which the authors note protects privacy and honesty [2, Section 3.3.1]. Gupta et al. catalog this instrument as 62 items [1, Tables 2, 7].

Validation of the questionnaire. Because no gold standard exists, the instrument was administered together with the department's existing interview-style questionnaire (which aggregates all classical textbooks) to 50 volunteers; Pearson correlations of percentage scores were r = 0.305 (p = 0.015) for Vata, r = 0.341 (p = 0.031) for Kapha, and r = 0.237 (p = 0.097, not significant) for Pitta. The authors treat this as adequate validation on the argument that r ≈ 0.30 is a medium effect size and that a stricter correlation is uninterpretable when the comparator itself lacks established accuracy - an argument they transparently flag as a limitation of CAM validation generally [2, Section 3.3.2].

Participants and classification. Of 150 randomly selected BAMS/MD-MS students (18-35 years), 90 met dual-type criteria: (1) secondary dosha score at least 50% of the primary, and (2) tertiary dosha at least 25% lower than the secondary. Ordering was collapsed (Vata-Pitta ≡ Pitta-Vata etc.), yielding Vata-Pitta n = 19 (4 + 15), Pitta-Kapha n = 50 (21 + 29), Vata-Kapha n = 21 (5 + 16); 61 men, 29 women; 72 aged 18-26 [2, Sections 3.3.3, 4]. All were clinically examined and declared healthy; ethics clearance and written consent were obtained. Screening found no participant scoring zero on any dosha and none with equal thirds - the minimum scores observed were 5% (Vata), 5% (Pitta), 20% (Kapha), and maxima 59% / 89% / 81% respectively [2, Section 4].

Physiological tests. In sessions ≥24 h apart (16:00-17:00, August-November), pulse rate, systolic and diastolic BP were recorded supine; on quick sitting and standing from lying and sitting (immediate + 2 min); during 3-minute isometric wall push; immediately, 2 and 5 minutes after the Harvard Step Test (20-inch bench, 30 steps/min, 5 min); cardiac efficiency index from pulse counts at 1-1.5, 2-2.5, 3-3.5 min post-exercise; and during a 2-minute cold pressor test (right hand, 4 °C water, readings every 30 s from the left arm). One-way ANOVA compared groups (SPSS 11.5) [2, Sections 3.4, 3.5].

D. Important findings

- **Null main result:** no parameter (HR, SBP, DBP across postural change, isometric exercise, cold pressor; cardiac efficiency index) varied significantly by dual-Prakriti group (Tables 2-6 of [2]). DR.
- **Single exception:** diastolic BP immediately after 5 min isotonic exercise differed significantly (p = 0.004, Table 7 of [2]); Vata-Kapha fell more (92.00 ± 7.211 mm Hg) than Vata-Pitta (97.79 ± 5.159) and Pitta-Kapha (97.12 ± 6.183). DR.
- **Interpretation offered:** greater vasodilation after isotonic exercise in Vata-Kapha may reflect lower Pitta-linked adrenergic (total peripheral resistance) drive; the authors link this to Pitta's tīkṣṇa (sharp) guṇa and to Udupa et al.'s catecholamine findings - explicitly a hypothesis, not a demonstrated mechanism. AI.
- **Population claim:** extreme single-dosha types are rare; every dosha contributes to every constitution to varying degree; predicting morbidity from Prakriti alone is difficult in dual types because the secondary dosha may nullify the primary's expression. AI (Discussion).

- **Questionnaire usability:** the self-scoring, single-textbook design is presented as a practical compromise addressing length, contradictory cross-textbook statements, and privacy. AI.

E. Limitations

The Pitta validation correlation was not statistically significant (r = 0.237, p = 0.097) [2, Section 3.3.2]. The questionnaire was validated against another unvalidated questionnaire. The sample is student-only, 18-35 years, single institution; dual types were collapsed pairwise (six nominal types into three groups) for statistical power. Season, age and psychological state are acknowledged confounders of constitution assessment. The single significant result (DBP after isotonic exercise) is one of many comparisons - the authors do not correct for multiplicity - so it should be read as hypothesis-generating. All participants being Ayurveda students (55 naive, 95 knowledgeable about Prakriti) could bias self-assessment.

F. Direct application to the project

Finding from paper	Information extracted	App module	How to use it	Evidence strength / limitation
Guṇa-based scoring model (Sec. 3.3.1)	120 points per guṇa split across features; maxima 960/600/1440; percentage = score ÷ max × 100 (absolute basis).	Prakriti scoring engine	Implement exactly this formula for the 62-question instrument; store per-dosha maxima in the rule schema so percentages are auditable; label scores "absolute % of that dosha's trait pool" in UI.	DR for the model's definition; its accuracy rests on criterion-level validation only.
Dvandvaja classification criteria (Sec. 3.3.3)	Secondary ≥ 50% of primary; tertiary ≥ 25% below secondary; pairwise order collapse.	Prakriti classification logic	Reuse as the platform's dual-dosha rule with published thresholds instead of inventing cutoffs; expose thresholds in the explanation panel ("why dual-Vata-Pitta").	AI/DR from one study; thresholds were study-design choices, not validated clinical cutoffs.
Validation statistics (Sec. 3.3.2)	r = 0.305 (V, p=.015), 0.341 (K, p=.031), 0.237 (P, p=.097, ns) vs departmental questionnaire, n = 50.	Evidence labelling; mentor reporting	Present the instrument as criterion-validated with explicit numbers; list Pitta ns as a known gap; do not call it "validated" without qualification.	DR; comparator questionnaire itself unvalidated.
Null cardiovascular result (Tables 2-6)	HR/SBP/DBP responses mostly do not differ by dual Prakriti type.	Wearable module framing	Supports the platform's honesty rule: do not map wearable cardiovascular signals to dosha type; treat [2] as direct negative evidence to cite.	DR; small dual-type sample (90), students, no multiplicity correction.
DBP-after-exercise exception (Table 7)	Vata-Kapha shows larger post-isotonic DBP fall (p = 0.004).	Future-research flag only	At most, document as a hypothesis for later cohort work; never surface to users as a dosha indicator.	DR of a single comparison; unadjusted for multiplicity; contradicts the general null pattern - report the tension openly.
Rare extreme types (Sec. 2, Discussion)	No zero or equal scores in 150 screened volunteers; extremes rare.	UX expectations; classification	Design the result screen for dual dominance as the normal case (5 of 6 type slots in practice); warn against promising "pure" Vata/Pitta/Kapha typing.	DR within sample (n = 150); generalises only as a tendency.

Table 4.3 Application mapping for Paper 2.

4.3 Paper 3 - Bhalerao & Patwardhan (2016): Good Reporting Practices for Prakriti Research

A. Paper information

- **Title:** Prakriti-based research: Good reporting practices.
- **Authors:** Supriya Bhalerao (BVDU, Pune), Kishor Patwardhan (BHU, Varanasi).

- **Journal/year:** Journal of Ayurveda and Integrative Medicine, 2016;7(1):69 (published 2016 May 24).
- **Access:** DOI 10.1016/j.jaim.2015.08.002; full text PMC4910291 (open access). Read in full.
- **Research type:** Methodological commentary / reporting checklist (no new data).

B. Research objective

As Prakriti links to genomics, biochemistry and physiology multiplied in the literature, the authors observed recurring protocol problems that make studies non-comparable. Their objective is an eight-domain reporting checklist that researchers and journal editors can enforce so that Prakriti-based findings become generalisable and reproducible [3, Abstract, Introduction]. For this project the paper functions as a specification source: it tells the platform exactly what metadata a Prakriti assessment must record to be scientifically defensible.

C. Methodology and content

The paper enumerates six common protocol problems (textbook discrepancies, ambiguous character assessment, inter-rater variability, scoring and weightage ambiguity, unclear final-type criteria, non-disclosure of the tool) and proposes eight reporting domains: (1) **Textual references** - name the source text; use terms like "Carakokta Prakriti"; justify any mixed tool given real contradictions (e.g., Charaka's small/short body for Vata vs Vagbhata's tall build; Sushruta's stealth tendency vs Charaka's normal psychosocial health; Charaka's minimal physical detail vs Ashtanga Hridaya's extensive physiognomy). (2) **Validation details** - report validation method; for research, ideally determine Prakriti two ways and include only concordant participants, or use two independent physicians; discuss self-report vs interviewer trade-offs. (3) **Assessment of characters** - operationalise every physical judgement (what counts as "round" eyes; how sun exposure is excluded from skin colour); state how undescribed characters were handled (e.g., body frame for Pitta). (4) **Scoring pattern** - declare absolute vs relative scoring: relative percentages use attained-sum denominators and do not measure expression against the dosha's full trait pool; absolute scoring does. (5) **Weightage assignment** - publish item weights and their rationale (stability, identifiability, dosha-specificity, guṇa basis). (6) **Final expression of type** - state the cutoffs separating ekadoshaja, dvandvaja and sama types; extreme single-dosha types are rare; publish every participant's three scores as supplementary data. (7) **Full tool disclosure** - publish the complete questionnaire and weights. (8) A summary table (Table 1) consolidates the checklist [3, Sections 2-9]. The classical anchors used are Caraka-saṃhitā Vimāna-sthāna 8.95, Sushruta-saṃhitā Śārīra-sthāna 4.65-76, and Aṣṭāṅga-hṛdaya Sūtra-sthāna 3.84-102 [3, References 21-23].

D. Important findings

- **Absoluteness matters:** the same "50% Pitta" number means different things under relative vs absolute scoring; Tripathi's model [2] is absolute, and the platform should say so. DR (conceptual distinction with worked definition, Section 5).
- **Cutoff transparency is a reporting requirement, not a detail:** no classical text fixes when a person is dvandvaja; every tool must publish its thresholds. DR (Section 7).
- **Dual determination recommended:** two methods or two raters, retaining concordant participants, is the paper's recommended research standard. AI/DR (Section 3).
- **Textual contradictions are documented** with named examples across Charaka, Sushruta and Vagbhata. DR (Section 2).

E. Limitations

It is a commentary: it provides no empirical test of whether following the checklist improves reproducibility, and its recommendations are proposals open to debate by design [3, Introduction, Conclusion]. It focuses exclusively on Prakriti determination, saying nothing about Vikriti instruments or wearables.

F. Direct application to the project

Finding from paper	Information extracted	App module	How to use it	Evidence strength / limitation
Eight-domain checklist (Table 1)	Textual base, validation, character assessment, scoring, weightage, type cutoffs, full tool disclosure.	Prakriti engine; project documentation	Adopt as the project's internal documentation standard; ship the 62-question instrument + weights + thresholds as a JSON "instrument pack" per domain 7.	DR as a published checklist; effectiveness not empirically tested.
Absolute vs relative scoring (Sec. 5)	Definitions of both scoring bases and why denominators differ.	Prakriti scoring engine; UI	State in-app that scores are absolute (score ÷ dosha maximum); keep the formula visible; never compare raw percentages across doshas as if same-scale.	DR (definitional); choice between bases remains debated.
Type cutoffs (Sec. 7)	Requirement to publish ekadoshaja/dvandvaja thresholds.	Prakriti classification logic	Use Tripathi's thresholds [2] and print them in the explanation panel; store thresholds as data, not code constants.	AI/DR; specific thresholds need platform-level validation.
Two-method determination (Sec. 3)	Concordance requirement for research participants.	Validation protocol (Ch. 15)	For any pilot study on platform data, require questionnaire-classification agreement (e.g., retest after ≥2 weeks) before analysis.	AI; a recommended practice, not an empirical result.
Character operationalisation (Sec. 4)	Examples of ambiguous traits (eye shape, skin colour, frame).	Questionnaire wording QA	Audit each of the 62 items for definitional ambiguity; add clarifying examples where the classical description is vague.	AI; supports item-writing QA.
Named classical anchors (Refs 21-23)	Ch.Vi. 8.95; Su.Sa.Sha. 4.65-76; A.Hr.Su. 3.84-102.	Knowledge database (Ch. 9-10)	Store as source metadata on every Prakriti item; enables the "source text" field of each recommendation record.	CT; verse ranges are as cited by [3].

Table 4.4 Application mapping for Paper 3.

4.4 Paper 4 - Mills et al. (2019): Vikriti Questionnaire and Psychological States

A. Paper information

- **Title:** Relationships among classifications of ayurvedic medicine diagnostics for imbalances and western measures of psychological states: An exploratory study.
- **Authors:** Paul J. Mills, Christine Tara Peterson, Kathleen L. Wilson, Meredith A. Pung, Sheila Patel, Lizabeth Weiss, Suhas G. Kshirsagar, Rudolph E. Tanzi, Deepak Chopra (UC San Diego; Chopra Center for Wellbeing).
- **Journal/year:** Journal of Ayurveda and Integrative Medicine, 2019;10(3):198-202 (online 2018; PMID 30385015).
- **Access:** DOI 10.1016/j.jaim.2018.02.001; full text PMC6822152 (open access). Read in full.
- **Research type:** Cross-sectional exploratory study (questionnaire correlations).

B. Research objective

The study asks whether self-reported vikṛti imbalances (current dosha disturbances) correlate with Western psychometric measures of psychological state. The hypothesis is that Western psychological diagnoses and descriptors correspond to specific dosha imbalances, which would support using vikṛti assessment as a mind-body wellbeing indicator in research and clinical settings [4, Abstract, Introduction]. For the platform, this is the core evidence base for the Vikriti module's questionnaire structure and for the interpretation ceiling of its outputs.

C. Methodology

Participants were 101 English-speaking adults aged 30-80 with no current major medical or mental illness (81 women, 20 men; mean age 53.9 ± 11.7 years), recruited via UC San Diego and the Chopra Center for Wellbeing. Vikriti was assessed with a 60-item questionnaire developed by physicians at the Chopra Center: it begins with a brief Prakriti assessment, then assesses current imbalance for each dosha; respondents rate symptom frequency as "very often", "sometimes" or "not often", scored 5, 3 and 1 respectively, so a higher vikṛti score indicates more imbalance of that dosha [4, Methods]. Western measures were the PROMIS Anxiety short form (8 items), the Center for Epidemiologic Studies Depression scale (CES-D, 20 items), the Perceived Stress Scale (PSS-10), the Mindful Attention Awareness Scale (MAAS, 15 items), the Rumination-Reflection Questionnaire (RRQ, 25 items) and the 84-item Ryff wellbeing scale. Analysis in SPSS used factor analysis to reduce each dosha's vikṛti scores to a single factor score, then multivariate general linear models with the other two dosha factors as covariates and age, gender and BMI as additional covariates [4, Methods, Results].

D. Important findings

Finding	Detail	Location / type
Vata imbalance associations	Higher Vata-imbalance scores associated with more anxiety ($p \leq 0.05$), more rumination ($p \leq 0.01$), lower mindfulness ($p \leq 0.05$), and lower overall psychological wellbeing/quality of life ($p \leq 0.01$).	Results; abstract. DR (cross-sectional associations)
Pitta imbalance associations	Pitta imbalance associated with poorer mood ($p \leq 0.01$), lower mindfulness ($p \leq 0.05$), more anxiety ($p \leq 0.05$), more perceived stress ($p \leq 0.05$).	Results. DR (cross-sectional associations)
Kapha imbalance associations	Kapha imbalance associated with more perceived stress ($p \leq 0.01$), more rumination ($p \leq 0.05$), less reflection ($p \leq 0.05$).	Results. DR (cross-sectional associations)
Instrument structure	60 items; 3-point frequency scale scored 5/3/1; starts with brief Prakriti assessment; higher score = more imbalance; single-factor structure per dosha (eigenvalues 1.42 / 1.44 / 1.48; variance explained 70.8% / 69.3% / 73.9%).	Methods; Results. DR
Conclusion offered	"Symptoms of mind-body imbalances in Ayurveda are differentially associated with western assessments of psychological states"; dosha assessment may help assess physical and emotional wellbeing in research/clinical settings.	Conclusion. AI

Table 4.5 Findings of Mills et al. [4] with locations and evidence type.

E. Limitations

The authors state two limitations: vikriti was assessed only by questionnaire (not by traditional methods such as pulse diagnosis), and participants were relatively healthy without overt disease [4, Limitations]. This report adds three structural cautions the team must respect: the design is cross-sectional, so no causal direction can be inferred (anxiety may elevate Vata-type symptoms or vice versa); the questionnaire itself has no published reliability/validity study - its psychometric credentials in this paper are limited to factor analytic coherence; and the sample (wellbeing-center affiliated, majority women, mean age ~54) limits generalisation. It is therefore not a universally validated diagnostic instrument, and the platform must not present it as one.

F. Direct application to the project

Finding from paper	Information extracted	App module	How to use it	Evidence strength / limitation
60-item structure + 5/3/1 scale	Frequency-anchored imbalance items per dosha; brief Prakriti prefix; higher = more imbalance.	Vikriti assessment engine	Adopt the response model (3-point frequency, 5/3/1) and per-dosha sub-scales; implement re-assessment prompts weekly/biweekly since states fluctuate; keep the brief Prakriti prefix only as a reminder, not a reclassification.	DR for structure; no independent validation study of the instrument exists.
Scoring approach	Sum per dosha; single-factor structure supported by factor analysis; no published clinical cutoffs.	Vikriti scoring	Report per-dosha imbalance scores as relative trends within the user's own history (higher/lower than their baseline), not as diagnostic levels; avoid inventing pathological cutoffs.	DR for factor structure; cutoffs absent - any thresholds are DP.
Psychological-state associations	Vata: anxiety, rumination, low mindfulness, low wellbeing; Pitta: mood, stress, anxiety; Kapha: stress, rumination, low reflection.	Interpretation layer; psychoeducation content	Generate educational copy ("higher Vata-imbalance scores in studies co-occurred with anxiety and rumination") with citation; never assert that the user has anxiety/depression.	DR (cross-sectional); correlation only - no causality, no diagnostic performance.
Factor analysis method	One-factor-per-dosha reduction; covariate adjustment (age, sex, BMI).	Analytics plan	Mirror the method in platform pilots: report per-dosha factor scores and adjust demographic covariates in any validation analysis.	DR (method); sample-specific.
Healthy sample caveat	No overt disease; self-report only.	Scope & disclaimers	State in-app that findings derive from generally healthy adults and are exploratory; route persistent high-distress scores to general wellbeing resources, not dosha advice.	DR (as stated by authors).

Table 4.6 Application mapping for Paper 4.

4.5 Paper 5 - Anagha & Deole (2021): Dinacharya, the Daily Regimen (CSNE Book Chapter)

A. Paper information

- **Title:** Daily regimen for preservation of health (Dinacharya).
- **Authors:** Anagha S., Deole Y.S.; reviewer/editor Basisht G.; Charak Samhita Research, Training and Skill Development Centre, IPGT&RA;, Jamnagar.
- **Publication/year:** Charak Samhita New Edition (CSNE), Health section, first published 2021 Feb 18; DOI 10.47468/CSNE.2021.e01.s09.044 (Crossref-registered book chapter, pp. 173).
- **Access:** Publisher's open site carakasamhitaonline.com (read in full); DOI resolves to the same record.
- **Research type:** Peer-reviewed narrative review of classical sources with contemporary research annotations.

B. Research objective

The chapter compiles the classically prescribed daily routine - waking, elimination, oral hygiene, eye/nose/ear care, massage, exercise, bathing, diet conduct, social conduct and sleep - and annotates each component with classical citations and selected contemporary research, with the stated premise that the routine must be adapted to constitution (deha prakṛti), strength (bala), age (vaya), season (ṛtu) and disease state [5, opening]. It is the backbone source for the platform's Dinacharya knowledge base.

C. Methodology and content structure

The chapter proceeds through the canonical sequence of daily activities and cites primary verses for each step; its "related chapters" note ties the content to Caraka-saṃhitā Sūtra-sthāna chapters 5 (Matrāśītīya), 6 (Tasyāśītīya - seasonal regimen), 7 (Naveganadharaṇīya) and 8 (Indriyopakramaṇīya - conduct) [5, More Information]. Contemporary evidence is cited inline as numbered references (e.g., studies on early risers' physical activity, tongue scraping and bacterial load, oil pulling and plaque, sesame-oil infant massage, immersion vs shower bathing, soap pH, coconut-oil protein loss).

D. Important findings

The chapter's real deliverable for this project is the citation-verified component catalogue in Table 4.7. Two features deserve emphasis. First, it contains a constitution-specific sub-table: chewing-stick selection by dominant dosha (Vata: sweet/astringent tastes - madhuka, khadira; Pitta: bitter - nimba; Kapha: pungent - karanja, arka), which is precisely the shape of rule the platform's engine needs (classical source + doṣa condition + item). Second, it quantifies classical limits where the classics do so (exercise to half strength; anjana at 5-8 day night intervals; water temperature by body region and season), giving implementable conditions rather than vague advice.

Component (Sanskrit)	English	Classical citation(s) via [5]	Key condition / limit	Modern note in [5]
Brahma-muhūrta utthāna	Waking before sunrise	Aṣṭāṅga-saṅgraha Sū. 3/2	~45-60 min before sunrise	Early waking correlates with higher physical activity in children; healthier adults prefer earlier wake times.
Śauca vidhi / vega	Elimination habits	Cha.Sa. Sū. 7/3-6	Do not suppress natural urges; morning urge signals gut health	Survey: once-daily defecation a minority pattern; early morning most common time.
Dantadhāvana	Brushing / chewing sticks	A.Hr.Sū. 2/2-3; Cha.Sa. Sū. 5/71-73; Su.Sa.Ci. 24/6; A.S.Sū. 3/5	9-inch fresh stems; taste by doṣa (Vata madhura/kaṣāya; Pitta tikta; Kapha katu); also after meals and before sleep	Chewing sticks show mechanical/chemical cleansing comparable or superior to toothbrushes.
Jihvā-nirlekhana	Tongue scraping	Cha.Sa. Sū. 5/76	Curved, non-sharp scraper; gold/silver/copper/tin/brass	Reduces bacterial load and plaque; plastic cleaners most effective in a three-type comparison.
Gandūṣa / kavala	Oil pulling	Cha.Sa. Sū. 5 (mouth-freshener and gargling section, cited by [5])	Sesame oil daily; gandūṣa = full hold, kavala = gargle volume	Reduced plaque index and gingival scores and aerobic colony counts in adolescents.
Añjana	Collyrium	Cha.Sa. Sū. 5/15-17	Cleansing (rasāñjana) only at night, every 5-8 days; never strong collyrium by day	Rasāñjana + honey relieved signs of mucopurulent conjunctivitis in a clinical study.
Nasya	Nasal oil instillation	Cha.Sa. Sū. 5/56-63	Aṇu taila; reaches śṛṅgāṭaka marma	Included in India's AYUSH self-care protocol during COVID-19.
Dhūmapāna	Medicated smoke	Cha.Sa. Sū. 5/34-35, 5/46-49	Eight prescribed times; after bath etc.; pacifies Vata-Kapha	Rarely practiced today; used mainly post-emesis.
Karṇapūraṇa	Ear oiling	A.Hr.Sū. 2/10; A.Hr.Sū. 22/32	Daily oil drops for hearing	Now mostly therapeutic, not daily.

Component (Sanskrit)	English	Classical citation(s) via [5]	Key condition / limit	Modern note in [5]
Abhyaṅga	Oil massage	A.Hr.Sū. 2/8	Daily; energises, nourishes, improves sleep	Sesame-oil massage in infants improved growth and sleep vs other oils.
Samvāhana / udvartana	Gentle / powder massage	Su.Sa.Ci. 24/83; A.Hr.Sū. 2/16	Samvāhana for Vata/thin build; udvartana (against hair direction) for Kapha/obesity	Udvartana studied in obesity and dyslipidaemia.
Vyāyāma	Exercise	A.Hr.Sū. 2/11-14	Limit: half of one's strength; more in winter/spring for rich diets; contraindicated in Vata-Pitta disorders, children, elderly; excess causes thirst, emaciation, dyspnoea, bleeding disorders	Regular moderate activity prevents lifestyle disease; dose matters like a drug.
Snāna	Bath	Cha.Sa. Sū. 25/40 (as cited by [5]); Su.Sa.Ci. 24/61	Hot water for body below neck, cold for head; seasonal caution (very cold winter water aggravates Vata-Kapha; very hot summer water aggravates Pitta)	Immersion bathing beat showers on fatigue/stress/pain in an RCT; soap pH matters (skin 5.4-5.9).
Āhāra vidhi	Eating rules	Cha.Sa. Vī. 1/21-25; Su.Sa.Ut. 34/75	Eat when hungry, previous meal digested; twice daily; no frequent eating	Contemporary nutrition parallels noted.
Sadvṛtta	Code of conduct	Cha.Sa. Sū. 8/25	Social/professional conduct prevents exogenous disease	Behavioural-health research gap flagged.
Nidrā	Sleep	Cha.Sa. Sū. 21/30 (as cited by [5])	Proper quantity and time; depends on age, constitution, occupation	Appropriate sleep linked to happiness, nourishment, strength, knowledge, longevity.

Table 4.7 Dinacharya component catalogue extracted from [5] (classical citations as given by the source).

E. Limitations

The chapter is a narrative review without systematic search or quality grading; its modern citations are selective annotations rather than evidence synthesis. Verse numbers are given per the editors' referenced editions and can differ across recensions (the source itself prints one apparent typographical variant in the gargling section). It addresses healthy-person routine; therapeutic adaptation is out of scope. For the platform this means: cite it for what the tradition prescribes and for the enumerated conditions, and route modern-evidence claims through the specific primary studies rather than through the review's summaries.

F. Direct application to the project

Finding from paper	Information extracted	App module	How to use it	Evidence strength / limitation
Full component catalogue with verses (Table 4.7 here)	16 dinacaryā components, Sanskrit names, conditions, doṣa-specific variants.	Ayurveda knowledge DB; recommendation engine	Seed the dinacaryā collection: each record = Sanskrit name + IAST + English + classical citation + condition + precaution + evidence tag CT (modern notes tagged separately).	CT; verse citations are as given by the peer-reviewed CSNE chapter.
Doṣa-specific tooth-stick table	Vata sweet/astringent; Pitta bitter; Kapha pungent, with named plants.	Recommendation rules (Ch. 14, Rule R1)	Template for constitution-conditional personalisation; implement as Prakriti-conditional rule with classical source field.	CT; classical recommendation, no modern trial of taste-based selection.
Exercise half-strength limit + contraindications	A.Hr.Sū. 2/11-14 conditions; excess-exercise harm list.	Vyāyāma rules; wearable gating	Rule: cap recommended intensity at self-reported half capacity; suppress intensity advice when Vikriti flags Vata/Pitta aggravation or user is elderly/underweight.	CT; aligns directionally with modern dose-response views cited in [5].
Bath water-temperature rules	Hot below neck, cold for head; seasonal cautions.	Dinacaryā rules; seasonal module	Season-conditional bathing advice with citations; straightforward low-risk rule set.	CT.
Eating rules	Eat on prior-meal digestion, twice daily, no grazing.	Āhāra rules	Meal-timing education content; combine with wearable meal-logging if present; do not convert into restrictive dieting.	CT.
Modern annotations (numbered refs)	Tongue scraping, oil pulling, massage, bathing studies.	Evidence tags in knowledge DB	Where a modern study is named in [5], record the primary citation and tag the claim as observational unless a trial is specified (e.g., immersion-bathing RCT).	Mixed; quality varies, verify each primary study before lifting claims.

Table 4.8 Application mapping for Paper 5.

4.6 Paper 6 - Dhiman et al. (2026): Ritucharya Critical Narrative Review (Abstract-Level Analysis)

A. Paper information

- **Title:** Ritucharya, the Ayurvedic Seasonal Regimen, for Health Maintenance and Disease Prevention: A Critical Narrative Review of Classical Rationale and Contemporary Biological Evidence.
- **Authors:** Diksha Dhiman, Neelam Gautam, Pawan Kumar Tripathi, Kanika Singhal, Ishika Bhatt, Rishabh Rawat.
- **Journal/year:** Asian Journal of Medicine and Health (ScienceDomain International), 2026;24(8):43-62; published 2026 Jul 30; ISSN 2456-8414.
- **Access status:** DOI 10.9734/ajmah/2026/v24i81409 resolves to the publisher site, which is protected by bot-verification; the full text could not be retrieved via any open archive during this research. **Full text was therefore NOT reviewed.** All statements below derive from the publisher-verified abstract and Crossref metadata.
- **Research type:** Critical narrative review (per title and abstract).

B-E. What the verified abstract establishes

The abstract states the review's method and conclusions in usable detail, and four elements are directly actionable for this project. First, it reconstructs the classical scheme as a coherent system built on the solar half-years (adaña-kāla / visarga-kāla), doshic accumulation and pacification, seasonal behaviour of digestive capacity, and the transitional vulnerability of inter-seasonal junctions (ṛtu-sandhi) [6, Abstract]. Second, it maps contemporary evidence onto three domains: chronobiological organisation of human physiology, including documented annual variation in immune and metabolic gene expression; seasonal restructuring of the gut microbiome linked to diet, cold and energy balance; and seasonality of cardiovascular and infectious disease [6, Abstract]. Third, it judges biological plausibility of a season-adaptive regimen to be "reasonably well supported by mechanistic and epidemiological work of moderate to high quality", while stating that direct clinical evidence for Ritucharya as an intervention "is, by contrast, sparse, methodologically weak and concentrated in lower-tier outlets", and that "constitution-based individualisation rests on preliminary and partly inconsistent data" [6, Abstract - quoted]. Fourth, it positions the regimen as "a biologically plausible and historically sophisticated preventive heuristic whose specific efficacy claims remain largely untested", and calls for rigorous prospective evaluation, standardised operationalisation and mechanistically anchored outcome measurement [6, Abstract - quoted].

This report treats those abstract sentences as the paper's contribution because nothing beyond them could be verified. Team members should obtain the full text through institutional access before the seasonal module is finalised; Chapter 15 lists this as an open validation item.

F. Direct application to the project

Finding from abstract	Information extracted	App module	How to use it	Evidence strength / limitation
Seasonal framework structure	Half-year kālas, doshic accumulation/pacification, agni seasonality, ṛtu-sandhi vulnerability.	Environmental/seasonal module	Model the season object with these four attributes; drive ṛtucaryā rules from season + Vikriti, never from wearable data.	AI per abstract; classical scheme, not an empirical result. Full text unverified.
Plausibility vs efficacy split	Biological plausibility reasonably supported; direct clinical efficacy sparse and weak.	Evidence labels on seasonal recommendations	Tag every seasonal recommendation "plausible, untested as intervention"; quote [6] in mentor documentation.	DR as the review's stated conclusion (abstract-level).
Constitution-based individualisation "preliminary"	Season × Prakriti personalisation lacks consistent evidence.	Recommendation engine caution	Present season-Prakriti combined advice as educational, with explicit uncertainty note; avoid implying proven personalisation.	DR as stated in abstract; aligns with this report's own null finding for dosha-linked physiology [2].
Cross-checkable external anchor	Documented annual variation in immune/metabolic gene expression.	Scientific context content	Cite Dopico et al. [21] directly (>4,000 seasonally expressed transcripts; pro-inflammatory profile in European winter) rather than relying on the review's summary.	DR in [21]; [6] names the domain but full text unavailable for its specific citations.

Table 4.9 Application mapping for Paper 6 (abstract-level).

4.7 Supplementary Sources Analysed for This Project

Beyond the six primary papers, the following verified sources contribute specific instruments or validation numbers. They are summarised here at the depth this report actually uses; full citations appear in the bibliography.

- **Mysore Tridosha Scale (Shilpa & Venkatesha Murthy 2011) [7].** A 157-item psychological self-report scale assessing the mental manifestation of the three doshas, developed and standardised with internal consistency,

split-half and test-retest reliability and content validity (Gupta et al. [1, Tables 2, 7] record validation samples 18 & 10 / 1,548 / 60; proprietary licence). Use: optional psychological extension of the Prakriti module; note licence and length before adopting.

- **Brief Prakriti Inventory (Bhargav et al. 2025) [8].** A 21-item self-report instrument built from classical texts; Multiple Correspondence Analysis, Latent Class Analysis and Item Response Theory in a community sample of N = 1,857; reliability 0.75-0.87 across doshas; test-retest ICCs 0.83-0.90; convergent validity r = 0.78-0.84 with AyuSoft; divergent validity against Western personality (Mini-IPIP). The strongest open, brief comparator for the platform's 62-question instrument.
- **Wrist-worn accuracy (Shcherbina et al. 2017) [10].** Sixty participants of diverse age, BMI, skin tone and fitness wore multiple devices against telemetry and indirect calorimetry. Median heart-rate error below 5% for six devices during cycling; no device achieved energy-expenditure error below 20%; error higher for darker skin tone, higher BMI, and walking. Use: justifies collecting HR but not calorie estimates; mandates diversity-aware framing.
- **Chest-strap RR quality (Gilgen-Ammann et al. 2019) [11].** Polar H10 chest strap achieved 99.6% RR-interval signal quality (99.4% at high intensity), beating a Holter reference (94.6%, 89.8% high intensity), r = 0.997. Use: shows HRV-grade interval data is feasible from PPG/ECG hardware - but smartwatch PPG is a different sensor class; treat watch HRV as trend data [12, 30].
- **HRV metrics primer (Shaffer & Ginsberg 2017) [12].** Defines RMSSD/SDNN, respiratory sinus arrhythmia and normal ranges; supports presenting HRV as a stress-recovery index, not a dosha measure.
- **Sleep technology position papers (de Zambotti et al. 2024 [13]; Chinoy et al. 2022 [14]; Lee et al. 2023 [15]; Stucky et al. 2021 [16]).** Home-condition validation of four trackers: epoch-by-epoch sleep sensitivity 0.93-0.96, wake specificity only 0.35-0.45, stage tracking "high variability" - authors advise using devices for sleep-wake outcomes, not stages [14]. Multicenter validation of 11 trackers vs polysomnography (349,114 epochs): macro-F1 0.26-0.69 by device and stage [15]. Fitbit Charge 2 in shift workers showed useful total-sleep-time estimates with stage limits [16]. Use: sleep duration = collect; sleep stages = collect-if-available with explicit uncertainty.
- **Wearable oximetry (Jiang et al. 2026) [17].** Under controlled hypoxia, Apple Watch Series 7 and Masimo MightySat both overestimated SpO_2; readings within ±2% in 32.14% vs 49.03% of measurements; root-mean-square error 5.82% vs 3.52% below 88% SpO_2. Use: SpO_2 is trend-grade on watches; do not display as clinical saturation.
- **Smartwatch population screening (Perez et al. 2019) [18].** Apple Heart Study: 419,297 participants; 0.52% received irregular-pulse notifications; positive predictive value 0.84 for simultaneous AF on ECG patch. Use: demonstrates large-scale PPG pulse analytics feasibility and the notification-ethics model the platform should imitate (conservative, opt-in, care-pathway aware).
- **Activity tracker meta-analysis (Chevance et al. 2022) [19].** Fitbit-class devices underestimate heart rate, steps and energy expenditure; average underestimation acceptable for steps and heart rate, not energy expenditure. Use: steps/active-minutes = reliable backbone variables.
- **Reporting quality of wearable research (Chan et al. 2022) [20].** Systematic review finds adherence and validity reporting inconsistent across medical studies of trackers. Use: the platform's own pilots should pre-register adherence/validity reporting.
- **Seasonal biology (Dopico et al. 2015) [21].** More than 4,000 protein-coding mRNAs in white blood cells and adipose tissue show seasonal expression, with inverted patterns between Europe and Oceania; a pro-inflammatory transcriptomic profile in European winter (higher soluble IL-6 receptor and C-reactive protein). Use: the strongest single external anchor for a season-adaptive routine module - and a reminder that seasonality is hemisphere- and geography-dependent.

- **AI/digital Prakriti review (Allam et al. 2026) [22]; Ayurveda preventive healthcare review (Rahiman et al. 2026) [23].** Recent reviews confirming standardisation, explainability and evidence integration as open problems - the gap the platform targets (Chapter 11).
- **Platform documentation [24, 25].** Health Connect data-type catalogue and permission model; HealthKit authorisation model - analysed in Chapter 8.

5. Prakriti Research Findings

This chapter consolidates what the reviewed literature supports about assessing baseline constitution, and converts it into engineering requirements for the Prakriti assessment engine. The evidence base is the scoping review [1], the questionnaire study [2], the reporting checklist [3], and the two modern instruments [7, 8].

5.1 What Prakriti Is, Operationally

Prakriti is the individual's relative dosha proportion at conception, classified classically into seven types (three single-dosha, three dual, one balanced) and further into ten by modern researchers who order the dual types [1, Introduction]. It is treated as lifelong-stable [1, 2]. Physiologically, Charaka's account assigns 8 guṇas to Vata, 5 to Pitta and 12 to Kapha, each producing observable physical, physiological and psychological features; this unequal attribute count is why percentage-of-maximum scoring exists at all [2, Section 3.3.1]. Classical assessment anchors are Caraka-saṃhitā Vimāna-sthāna 8.95, Sushruta-saṃhitā Śārīra-sthāna 4.65-76 and Aṣṭāṅga-hṛdaya Sūtra-sthāna 3.84-102, which disagree on some attributes [3, Sections 2, References 21-23].

5.2 Assessment Approaches and Their Evidence

Three tool families exist. Questionnaires dominate (16 of 32 reviewed studies) and are the most validated, yet only 3 of 16 were validated, 3 carried both reliability and validity, and AyuSoft leads [1]. Algorithmic/ML approaches (5 studies) label data by physicians or questionnaires and inherit their weaknesses; performance metrics exist but datasets and code are rarely open [1, Tables 3-4]. Device approaches (11 studies, mostly pulse-waveform based, e.g., Naḍī Taranginī-class instruments) are the most objective in principle but none is adequately validated with accessible data [1, Tables 5-6]. For a student platform, the questionnaire route is the only evidence-supported option today; devices and ML are future-research items, a conclusion identical to the review's own [1, Conclusion].

5.3 The Scoring Model the Platform Should Implement

- **Instrument:** the 62-question self-assessment based on Charaka (via Patwardhan-Sharma, as used by [2]; catalogued in [1, Tables 2, 7]).
- **Scoring:** each guṇa carries 120 points distributed equally among its observable features; dosha maximum = 120 × guṇa-count (Vata 960, Pitta 600, Kapha 1440); user percentage = points earned ÷ dosha maximum × 100 (absolute basis) [2, Section 3.3.1].
- **Classification:** primary dosha = highest percentage; dual (dvandvaja) when secondary ≥ 50% of primary and tertiary ≥ 25% below secondary; order-collapsed dual labels (e.g., Vata-Pitta ≡ Pitta-Vata) [2, Section 3.3.3].
- **Stability framing:** every item must ask about lifelong tendencies, following the 8-of-16 questionnaire practice noted by [1, Section 3.1.1] and the reporting requirements of [3, Sections 3-8].
- **Disclosed validation:** criterion-level correlations r = 0.305 (Vata, p = .015), 0.341 (Kapha, p = .031), 0.237 (Pitta, p = .097 ns) against a departmental instrument, n = 50 [2, Section 3.3.2] - report as-is, never as "fully validated".

Two implementation safeguards follow from the evidence. First, because physician inter-rater reliability is only 0.20-0.40 [1, Section 4], the platform should not aspire to "replace an Ayurvedic physician" in its copy; the honest claim is structured self-assessment with disclosed limitations. Second, because extreme single-dosha types are rare (zero zero-scores and zero equal-scores in 150 screened volunteers [2, Section 4]), the result UX must treat dual constitutions as the normal case and balanced (sama) types as explicitly rare, with the balanced case handled by a dedicated rule branch rather than a dosha-specific one.

5.4 Psychometric Targets for the Platform's Own Validation

The platform can set quantified targets from the reviewed instruments: internal consistency ≥ 0.75 per dosha subscale and test-retest ICC ≥ 0.83 (BPI achievements [8]); inter-rater agreement documentation if practitioner review is ever added (AyuSoft's IRR precedent [1, Table 7]); and convergent correlation ≥ 0.78 against an established instrument (BPI vs AyuSoft [8]) if licensing or a comparable open instrument is available. Chapter 15 turns these into a concrete validation protocol.

6. Vikriti Research Findings

Vikriti is the current dosha state - what is disturbed now, not who the person is. The reviewed evidence for assessing it comes from one peer-reviewed instrument study [4], supported by the review's scoping statement that Vikriti is a distinct, fluctuating construct [1].

6.1 What Vikriti Is, Operationally

Vikriti reflects deviations from the person's own Prakriti caused by diet, season, stress and behaviour [1, Introduction; 4]. Operationally for the app: a Vikriti assessment compares current-state answers against the user's baseline, producing per-dosha imbalance scores. The Mills et al. questionnaire operationalises "current" as symptom frequency ("very often" / "sometimes" / "not often", scored 5/3/1), with higher totals meaning more imbalance of that dosha [4, Methods].

6.2 The 60-Item Questionnaire: Structure, Scoring, Domains, Limits

- **Structure:** 60 items; begins with a brief Prakriti assessment, then assesses current imbalance per dosha; developed by physicians at the Chopra Center for Wellbeing [4, Methods].
- **Response scale:** 3-point frequency (5/3/1). Scoring: sum per dosha; higher = more imbalance. No diagnostic cutoffs are published [4, Methods, Results].
- **Domains:** symptom clusters per dosha reflecting classical imbalance pictures (the paper reports the three scales as coherent single factors: eigenvalues 1.42/1.44/1.48; variance 70.8%/69.3%/73.9%) [4, Results].
- **Validation status:** exploratory; cross-sectional associations with psychological scales in 101 healthy adults; factor coherence demonstrated; no reliability study, no criterion standard, no clinical cutoffs [4, Results, Limitations].
- **Stated limitations:** questionnaire-only assessment (no pulse-diagnosis comparison); relatively healthy sample [4, Limitations].

6.3 What the Platform May and May Not Claim

The demonstrated associations give the platform legitimate educational content: in this study, Vata-imbalance scores co-occurred with higher anxiety and rumination and lower mindfulness and wellbeing; Pitta with poorer mood, anxiety and stress; Kapha with stress, rumination and lower reflection [4, Results]. The platform may display such statements as study findings with citations. It may not: diagnose anxiety/depression from a Vikriti score; treat the score as a medical risk indicator; or imply that lowering a score treats a condition. The honest framing is "self-reported state assessment with research-documented associations, exploratory in nature".

6.4 Assessment Frequency and State Confounders

Because Vikriti is state-like, the assessment must be repeatable and short enough to retake; the platform should support weekly or biweekly re-assessment and after notable events (travel, season change, illness recovery). Confounders the UI should capture alongside scores include recent illness, travel/sleep disruption, acute stress, and (for women) menstrual-cycle phase, since none of the reviewed studies disentangles these from dosha-state changes - a gap the platform should acknowledge rather than absorb silently. Acute physiological context from wearables (e.g., a short-sleep week) belongs in the trend panel next to Vikriti scores, not inside them (Chapters 8, 14).

7. Prakriti vs Vikriti Comparison

The two constructs answer different questions - "who is this person?" versus "what is disturbed right now?" - and the platform must keep them separate end-to-end: separate instruments, separate scoring engines, separate database collections, separate UI surfaces. The comparative table below fills the exact structure requested by the project brief.

Feature	Prakriti	Vikriti
Meaning	Baseline constitution: relative Vata/Pitta/Kapha proportion at conception; believed lifelong-stable [1, 2].	Current dosha state/imbalance: fluctuating deviation shaped by diet, season, stress, behaviour [1, 4].
Purpose	Personalisation anchor: long-term routine, diet tendencies, exercise temperament, explanation of stable traits.	Adaptive signal: what to soften or support now; drives time-limited recommendations and reassessment prompts.
Stability	Stable across lifespan (conceptual claim of Ayurveda; no longitudinal instrument-retest literature found in reviewed sources) [1, 2].	Explicitly variable; re-assessment on the order of days-to-weeks is appropriate [1, 4].
Assessment approach	Structured self-report questionnaire on lifelong traits; classical examination by practitioners; experimental pulse devices/ML (unvalidated) [1, 2].	Structured self-report on current symptom frequency; no validated device or ML analogue found in reviewed sources [4].
Relevant questionnaire	62-item Charaka-based self-assessment [2; 1, Tables 2, 7]; comparators: AyuSoft (85), PPAT (54), BPI (21), Mysore Tridosha (157) [1, 7, 8].	60-item Chopra Center vikṛti questionnaire [4]. No other peer-reviewed vikṛti instrument was located in the reviewed set.
Dosha scoring	120 points/guṇa; % of dosha maximum (absolute basis): 960/600/1440 maxima [2].	Per-item 5/3/1 frequency points summed per dosha; higher = more imbalance; no published cutoffs [4].
Variables	Lifelong physical (frame, skin, hair, eyes), physiological (appetite, digestion, sleep pattern), psychological (memory, temperament) traits [1-3].	Current symptom frequency per dosha cluster; contextual confounders (stress, sleep, travel, cycle phase) recorded alongside [4].
Interpretation	Dominant or dual type with published thresholds (secondary ≥ 50% of primary; tertiary ≥ 25% below secondary) [2]; order-collapsed dual labels; balanced type rare.	Relative imbalance scores interpreted against the user's own baseline and history; research-documented associations with psychological states; no diagnosis [4].
Research limitations	No gold standard; physician IRR 0.20-0.40; criterion-level validation only (Pitta ns); field-level construct ambiguity [1-3].	Single exploratory study; instrument lacks standalone validation; healthy sample; cross-sectional correlations only [4].

Feature	Prakriti	Vikriti
App implementation	Onboarding module; one-time (with optional confirm-retest); result locked into user profile; drives recommendation context everywhere.	Repeating module; drives adaptive rules and reassessment nudges; never merged with Prakriti into one score; displayed next to wearable trends.

Table 7.1 Prakriti vs Vikriti - comparative analysis for implementation.

Handling of dual and balanced constitutions deserves one further note. The reviewed sources give thresholds for dual classification [2] but no threshold for "balanced" (sama) Prakriti; the honest implementation is to define balanced as a rare fallback case (e.g., all three doshas within a narrow band) with its own neutral routine set, and to disclose that the definition is a design decision, not a research finding. Similarly, Vikriti balance means "no dominant current imbalance", which the engine treats as "maintain dinacaryā, no corrective emphasis".

8. Smartwatch Integration Research

This chapter answers four questions: which wearable variables are worth collecting, what validation evidence supports each, how to access them on Android and iOS, and what the app may and may not do with them. The governing constraint, established by the null and near-null findings of [2] and confirmed by the absence of any contrary evidence in this review, is: wearable data is wellness context, never a dosha measurement.

8.1 Variable-by-Variable Evidence Table

Variable	What it measures / unit	Source & frequency	Research evidence	Potential app use + limitations
Heart rate (HR)	Instantaneous pulse rate, bpm; PPG optical or ECG electrical.	Watch all day; spot readings; workout streams [24, 25].	Median error < 5% during cycling for 6 of devices tested; error rises walking, higher BMI, darker skin tone; no device below 20% error for calories [10]; Fitbit-class underestimation acceptable on average for HR and steps [19].	Daily average and workout context. Reliable enough for trend display; do not build clinical thresholds. Display "trend" not "truth".
Resting heart rate (RHR)	HR at rest, bpm; device-estimated from lowest sustained values / sleep.	Daily estimate [24].	Derived from PPG HR which is acceptably accurate at rest [10, 19]; no review found a dosha correlation study.	Long-term stress/fitness trend line. Useful weekly trend; single-day changes are noisy.
HRV (RMSSD)	Beat-to-beat variability, ms; RMSSD is the standard short-term index [12].	Spot readings or overnight averages where supported [24, 25].	Chest-strap RR quality 99.6% shows interval data feasibility [11]; watch PPG HRV is noisier and position-sensitive vs ECG [12, 30]; no dosha mapping exists.	Stress-recovery context: nightly trend + weekly baseline; pair with Vikriti re-assessment. Never interpret alone; expect missing days.
Steps	Step count per day, count.	Continuous, passive [24, 25].	Best-validated wearable variable; average underestimation acceptable [19].	Backbone activity variable; feeds vyāyāma context and weekly goal logic. High confidence; still degrade gracefully if absent.
Active minutes	Minutes at moderate/vigorous effort, min; estimated from HR+motion.	Derived daily [24, 25].	Definition varies by vendor; HR-linked estimate inherits HR accuracy [10, 19].	Exercise-regimen context (classical vyāyāma dose vs modern guideline framing). Label as estimate; vendor formulas differ.
Sleep duration	Time asleep, hours/minutes; interval-based estimate.	Nightly, automatic [24, 25].	Home validation: TST/efficiency tracked well; sleep sensitivity 0.93-0.96, wake specificity 0.35-0.45 [14]; multicenter macro-F1 spread large [15].	Nidrā module: duration trend vs classical regularity advice. Strong for duration, weak for wake detection.

Variable	What it measures / unit	Source & frequency	Research evidence	Potential app use + limitations
Sleep stages	Awake/Light/Deep/REM classification per epoch.	Nightly if device supports [24, 25].	Stage classification remains the weak point: macro-F1 0.26-0.69 across 11 devices [15]; "best utilised for sleep-wake outcomes, not stages" [14].	Collect if available, display with uncertainty chip ("approximate"); never claim diagnostic sleep staging.
SpO_2 (oxygen saturation)	Peripheral oxygen saturation, %.	Spot or nightly if supported [24, 25].	Watches overestimate SpO_2; within ±2% in 32% of readings vs 49% for a reference oximeter; error grows under hypoxia (Arms 5.82%) [17].	Optional trend only. Given overestimation, suppress value display; if shown, add "not a medical measurement" tag. Consider de-prioritising for prototype.
Skin temperature	Peripheral/skin temperature, °C; nightly deltas vs personal baseline.	Nightly estimate if supported [24].	Health Connect exposes SkinTemperatureRecord deltas [24]; validation literature reviewed here is thinner; no dosha studies.	Future enhancement: circadian/seasonal context. Mark as exploratory; require baseline accumulation before showing anything.
Respiratory rate	Breaths per minute, estimate.	Nightly where supported [24].	Device-estimated; no dedicated validation in reviewed set; treat like HRV-grade trend.	Future enhancement next to HRV; same uncertainty handling.

Table 8.1 Wearable variables: meaning, evidence and intended app use (units: bpm = beats per minute; ms = milliseconds).

8.2 Directly Measured vs Estimated - A Non-Negotiable Distinction

Optical (PPG) heart rate is a measured pulse waveform whose rate is computed; resting heart rate, HRV summaries, active minutes, sleep duration and stages, SpO_2 and skin-temperature deltas are algorithmic estimates from underlying signals, and vendor algorithms differ and change with firmware [10, 14, 15, 19, 24, 25]. The platform's data model should therefore carry a provenance flag per reading (measured / vendor-estimated) and the UI must label estimates. This is also an equity issue documented by validation research - accuracy degrades with darker skin tone and higher BMI [10, 15] - which the app should disclose in its data-quality note rather than hide.

8.3 Platform Access: Android Health Connect

- **Model:** Health Connect is an on-device central store for health and fitness data; apps declare record-level permissions (READ_/WRITE_) in the manifest and request user authorisation; Google Play requires a health-apps declaration for access to sensitive types [24].
- **Relevant record types (verified in the official data-types catalogue):** HeartRateRecord (series, bpm), RestingHeartRateRecord, HeartRateVariabilityRmssdRecord (ms), StepsRecord (count), SleepSessionRecord (interval with stage list), OxygenSaturationRecord (%), SkinTemperatureRecord (series of deltas), RespiratoryRateRecord, plus ExerciseSessionRecord, BodyMeasurement and Nutrition records [24].
- **Special permissions:** background reads require READ_HEALTH_DATA_IN_BACKGROUND; reads older than 30 days require READ_HEALTH_DATA_HISTORY - both declared separately from type permissions [24].
- **Operational cautions from the documentation:** rate limits apply to reads; aggregation APIs exist for daily sums (steps, duration); historical availability depends entirely on what other apps wrote; Google Fit APIs are scheduled for deprecation (supported until end of 2026) so Health Connect is the correct Android target [24].

8.4 Platform Access: Apple HealthKit

- **Model:** HealthKit is the iOS central health repository; apps request per-type authorisation to read and to share; the user can change permission per app at any time in the Health app, and apps must handle data changes occurring outside the app [25].

- **Relevant types:** heart rate, resting heart rate, heart-rate variability (SDNN on Apple Watch), step count, sleep stages (since watchOS 9), oxygen saturation, wrist temperature deltas, respiratory rate - all user-permissioned [25].
- **Privacy requirements:** health data is sensitive; apps must request permission for any read or write and protect data at all times; Apple reviews health-app data use [25].
- **Multi-source reality:** HealthKit merges data from watch, iPhone and third-party apps with source attribution - the platform should record the source device per sample when interpreting quality [25].

8.5 Synchronization, Missing Data, Timestamps and Consent

Practical engineering rules follow from the platforms and the validation literature. Sync: Health Connect and HealthKit are pull-based stores; schedule periodic reads (e.g., on app open and daily background job where permission allows) and keep an ingestion log with per-record timestamps; expect late-arriving backfill when the watch re-syncs. Missing readings: absence of a record type is normal (device capability, permission revoked, sensor off-wrist); the dashboard must show coverage per variable rather than pretending completeness. Timestamps and units: both platforms carry explicit start/end times and typed units; normalise to UTC internally, display in user-local time, convert bpm/ms/count/% per the record schema, and never recompute vendor estimates. Consent and privacy: request minimum scopes at first use with plain-language purpose strings; allow partial consent (user may deny sleep but grant steps); store no health data off-device beyond what features require; provide in-app data deletion; and never send wearable data to third parties - consistent with the platforms' own review requirements [24, 25].

8.6 What the App Must Not Do

No reviewed study validates any wearable variable as a doṣa indicator; the only dosha-cardiovascular study found mostly null group differences [2]. Therefore: the app must not label HRV/sleep/HR readouts with dosha names, must not claim "your Vata is high" from a low HRV, and must not generate alerts implying medical detection (the Apple Heart Study's conservative notification design - 0.52% notified, with a confirmatory pathway [18] - is the appropriate model for any future alerting). Framing language should be: "objective context that may relate to how you feel", paired with the user's own Vikriti self-report.

9. Ayurveda Knowledge Extraction

This chapter organises the recommendation universe into the six categories requested - dinacaryā, ṛtucaryā, āhāra, vyāyāma, nidrā, sadvṛtta - plus a cross-cutting seventh. Every row carries: Sanskrit name (transliterated), English meaning, explanation, classical source, research paper where applicable, relevant context, a possible app rule, and precautions. Evidence labels: CT = classical/traditional; DR = demonstrated research result; AI = author interpretation; DP = design proposal. The tables deliberately separate what tradition prescribes from what modern research supports, per the accuracy rules of this project.

9.1 Dinacharya (dinacaryā) - Daily Routine

The full component catalogue with classical citations was extracted in Table 4.7 from the peer-reviewed CSNE chapter [5]; that table is the canonical source for this category and is not repeated here. The engine-relevant rules distilled from it: wake-time regularity (brahma-muhūrta framing, adapted to the user's schedule as a regularity rule, DP); elimination regularity and urge-respect (Cha.Sa. Sū. 7/3-6); oral hygiene stack (brushing - taste-conditional by Prakriti; tongue scraping; oil pulling); sensory-care stack (añjana/nasya/karṇapūraṇa as optional classical items with their interval and time-of-day conditions); abhyaṅga before bathing; vyāyāma at half capacity; snāna with region- and season-conditional water temperature; and conduct rules during work hours.

9.2 Ritucharya (ṛtucaryā) - Seasonal Routine

The classical scheme (per the verified abstract of [6] and the related-chapters note in [5]): the year divides into two solar half-years - adaña-kāla (northern solstice half, ascending heat) and visarga-kāla (southern solstice half, ascending coolness) - with three seasons each; doṣas accumulate and pacify in season-ordered patterns; digestive strength (agni) varies seasonally (strong in cold seasons, weak in early winter-to-summer transitions per the scheme); and ṛtu-sandhi (seasonal junctions) are periods of transition vulnerability where the previous regimen is tapered and the next introduced gradually [6, Abstract]. Chapter 6 (Tasyāśītīya) of Caraka-saṃhitā Sūtra-sthāna is the classical anchor named in [5, More Information]. Contemporary plausibility anchors verified in this review: >4,000 seasonally expressed human transcripts with hemisphere-inverted patterns and a pro-inflammatory winter profile [21]. The engine's seasonal module should therefore: compute the user's season from geography and hemisphere; apply taper/transition logic at ṛtu-sandhi (about the junction window, DP); expose seasonal recommendations tagged CT with the [6]-level honesty label "biologically plausible; not yet tested as an intervention"; and never personalise season advice with wearable data.

Element	Classical content (as summarised in [6])	Modern anchor	Possible app rule + precautions
Adaña-kāla	Solar half-year with ascending heat/dryness; three seasons within it.	Seasonality of physiology documented [21].	Progressively lighter/cooling routine emphasis across the half-year; precaution: geography-specific adoption (hemisphere inversion shown in [21]).
Visarga-kāla	Solar half-year with ascending coolness/moistness; three seasons.	Pro-inflammatory winter profile in Europe [21].	Nourishing/routine-stabilising emphasis; precaution: not a treatment claim; label plausible-untested per [6].
Doṣa accumulation /pacification	Season-ordered accumulation and pacification of doṣas across the year.	No modern verification reviewed.	Display as educational calendar; do not compute "dosha load" from data; precaution: avoid implying measurable dosha kinetics.
Agni (digestive capacity) seasonality	Digestive strength varies by season in the classical scheme.	Seasonal metabolism variation discussed in [6] (abstract-level).	Adjust meal-regularity advice seasonally (lighter in peak heat); precaution: no clinical claims; keep general.
Ṛtu-sandhi	Junction periods between seasons; gradual regimen transition.	None reviewed.	Rule: during junction window, blend previous and next season advice; taper intensity changes; precaution: window length is a design choice (DP).

Table 9.1 Seasonal regimen knowledge structure (classical scheme per [6, Abstract]; biological anchors [21]; labels CT unless noted).

9.3 Ahara (āhāra) - Food and Eating Habits

The platform's food content must stay educational and non-prescriptive: no therapeutic diets, no herbs-as-medicine, per the project's own scope rules. The verified classical anchors from [5] are behavioural rather than compositional: eat when the previous meal is digested and on genuine hunger (Su.Sa.Ut. 34/75); twice-daily meal frequency without grazing; the āhāra-vidhi eating-rule set of Caraka-saṃhitā Vimāna-sthāna 1.21-25 (the classical "how to eat" rules - eat warm/unctuous in due quantity, etc.); and the taste-conditional logic demonstrated in the tooth-stick table (Vata-sweet/astringent, Pitta-bitter, Kapha-pungent) which generalises as a taste-education pattern, not as diet prescriptions [5]. App rules: meal-regularity tracking with user-logged meals; hunger-check prompt before meals (educational); season-conditional eating-pattern emphasis per 9.2; Prakriti-conditional taste-education content. Precautions: never restrict food groups; explicitly opt out for users with any disordered-eating flag; no weight-loss framing.

9.4 Vyayama (vyāyāma) - Exercise

Classical anchors: regular exercise lightens the body, strengthens digestion and reduces obesity; the dose is capped at half of one's strength; season and diet matter (those on rich diets should exercise especially in winter and spring); contraindications include Vata/Pitta disorders, childhood and old age; excess causes thirst, emaciation, breathlessness, bleeding disorders, fatigue, cough, fever and vomiting; post-exercise the whole body should be massaged comfortably (A.Hr.Sū. 2/11-14, per [5]). Modern anchors: regular moderate activity prevents lifestyle disease with dose-response behaviour [5, ref 23]; wearable-derived active minutes and steps are the platform's objective dose measures, with acceptable average accuracy for steps and HR [19]. App rules: personal "half-strength" calibration via a short perceived-exertion check (DP); season-conditional intensity (higher in cool seasons per classical text, aligned with modern outdoor-activity practicality); suppress intensity increases when Vikriti flags Vata/Pitta imbalance, when sleep debt is high, or when RHR trend is elevated; post-exercise recovery prompt (abhyaṅga framing). Precautions: no max-HR prescriptions; no exercise plans for users with cardiac flags without professional consultation notice; classical "half strength" is CT, not a physiologically validated threshold - label it as such.

9.5 Nidra (nidrā) - Sleep

Classical anchors: proper sleep in proper quantity and time supports happiness, nourishment, strength, fertility, knowledge and longevity; sleep needs vary by age, constitution and occupation (Cha.Sa. Sū. 21/30 as cited in [5]); abhyaṅga is described as sleep-promoting (A.Hr.Sū. 2/8, per [5]); bath before sleep and regular timing appear in the dinacaryā sequence [5]. Modern anchors: consumer trackers estimate duration well but wake/stage detection poorly [14, 15]; regularity is a usable, measurable property (bedtime/waketime consistency). App rules: nightly sleep-duration trend with 7-day consistency view (DP); bedtime-regularity nudges framed classically ("regular time of sleep is prescribed") and modernly; display tracker stages with uncertainty chip or omit if unsupported; optional evening-routine stack (bath, massage, screen-down wind-down) drawn from dinacaryā items with CT tags. Precautions: never diagnose sleep apnoea or insomnia; persistent complaints route to professional-care suggestion; do not use SpO_2 for sleep-disorder hints given its overestimation bias [17].

9.6 Sadvritta (sadvṛtta) - Conduct and Mental Wellness

Classical anchor: the code of right conduct prescribed for social and professional life prevents exogenous disease (Cha.Sa. Sū. 8/25, per [5]); the CSNE chapter links this domain to psychological and behavioural health and flags it as a research frontier [5]. Modern support in the reviewed set is indirect: the Vikriti study's psychological correlations [4] and the mindfulness/stress measures it used (MAAS, PSS) suggest which wellbeing constructs the platform can reference educationally. App rules: a small set of conduct/wellbeing practices with weekly check-ins (e.g., regular routine adherence, measured speech and sensory-rest breaks, gratitude/reflective journaling framed via RRQ's reflection construct - clearly labelled as modern instruments, not classical prescriptions); Vikriti-informed stress-education content citing [4]. Precautions: no mental-health diagnosis; escalation path for distress indicators; keep spiritual framing optional and non-sectarian.

9.7 Cross-Cutting: Dinacharya's Constitution-Conditional Pattern

One structural finding from [5] deserves its own heading because it is the template for the whole knowledge base: the tooth-stick table maps Prakriti → taste category → item list, with citations. The same pattern recurs (samvāhana for Vata/thin build vs udvartana for Kapha/obese; bath temperature by season and body region; exercise seasonality). The knowledge database should therefore implement a first-class "condition" field per record supporting Prakriti conditions, Vikriti conditions, season conditions and combinations - exactly the schema proposed in Chapter 12.

10. Sanskrit Terminology and Recommendation Database

This chapter fixes the Sanskrit vocabulary the platform will use, with transliteration and meaning, and defines the record schema of the recommendation database. Devanagari renderings of the six category anchors are shown in the terminology plate below (Figure 10.1) because correct Devanagari shaping requires a dedicated rendering pass; everywhere in running text this report uses IAST transliteration, which is the academically standard form (dinacaryā, ṛtucaryā, āhāra, vyāyāma, nidrā, sadvṛtta, doṣa, guṇa, agni, abhyaṅga, snāna, prakṛti, vikṛti).

Core Sanskrit Terminology

Devanagari, IAST transliteration and English meaning for the platform's routine categories and assessment terms

Devanagari	IAST	English meaning
प्रकृति	*prakṛti*	Prakriti - baseline constitution
विकृति	*vikṛti*	Vikriti - current imbalance state
दोष	*doṣa*	dosha - Vata / Pitta / Kapha principle
दिनचर्या	*dinacaryā*	Dinacharya - daily routine
ऋतुचर्या	*ṛtucaryā*	Ritucharya - seasonal routine
आहार	*āhāra*	Ahara - food and eating habits
व्यायाम	*vyāyāma*	Vyayama - physical exercise

Devanagari	IAST	English meaning
निद्रा	*nidrā*	Nidra - sleep
सद्वृत्त	*sadvṛtta*	Sadvritta - code of right conduct
अग्नि	*agni*	digestive capacity
अभ्यङ्ग	*abhyaṅga*	oil massage
स्नान	*snāna*	bathing
ब्रह्ममुहूर्त	*brahma-muhūrta*	pre-dawn window (~45-60 min before sunrise)
ऋतुसन्धि	*ṛtu-sandhi*	seasonal junction period

Figure 10.1 Core Sanskrit terminology plate: Devanagari, IAST transliteration and English meaning for the six routine categories plus key assessment terms.

Term (IAST)	Common spelling	Meaning in this platform's copy	Primary source
prakṛti	Prakriti	Baseline constitution (who the person is).	[1-3]

Term (IAST)	Common spelling	Meaning in this platform's copy	Primary source
vikṛti	Vikriti	Current imbalance state (what is disturbed now).	[1, 4]
doṣa	dosha	Vata / Pitta / Kapha regulatory principles.	[1, 2, 26]
guṇa	guna	Attribute of a dosha; the scoring unit of the 62-item instrument.	[2, 3]
dinacaryā	Dinacharya	Daily routine regimen.	[5]
ṛtucaryā	Ritucharya	Seasonal routine regimen.	[5, 6]
āhāra / āhāra-vidhi	Ahara	Food and the rules of eating.	[5, 26]
vyāyāma	Vyayama	Physical exercise.	[5]
nidrā	Nidra	Sleep.	[5, 26]
sadvṛtta	Sadvritta	Code of right conduct.	[5, 26]
agni	Agni	Digestive capacity (contextual in seasonal advice).	[5, 6]
abhyaṅga	Abhyanga	Oil massage.	[5]
snāna	Snana	Bathing.	[5]
brahma-muhūrta	Brahma muhurta	Pre-dawn window (~45-60 min before sunrise).	[5]
ṛtu-sandhi	Ritu sandhi	Seasonal junction period.	[6]
dvandvaja / ekadoshaja / sama	as transliterated	Dual-dosha / single-dosha-dominant / balanced constitution.	[2, 3]

Table 10.1 Terminology register for consistent use across UI, rules and documentation.

10.1 Recommendation Record Schema (Conceptual)

Every recommendation in the knowledge database is one record with the fields the project brief requires: sanskrit_name, transliteration, english_name, explanation, ayurvedic_principle, classical_source (text + chapter/verse as cited by the reviewed modern source), research_reference (one of the bibliography keys or null), relevant_context (prakriti / vikriti / season conditions), suitability_conditions, precautions, evidence_level (one of CT / DR / AI / DP), and implementation_rule (pointer to a rule id). The JSON shape proposed for the FastAPI/JSON-rules stack is shown in Chapter 12.4. Keeping classical_source and research_reference as separate fields is what allows the UI to display "classical basis" and "research status" side by side without ever blending them.

11. Research Gap and Novelty Analysis

11.1 What Already Exists

Prakriti assessment systems: sixteen questionnaire tools (up to 157 items, variously validated), five ML classifier studies and eleven pulse-device prototypes are catalogued by [1]; the best-validated is AyuSoft (proprietary), with PPAT open-source and the BPI a brief open comparators [1, 8]. Vikriti assessment systems: the 60-item Chopra Center questionnaire is the principal peer-reviewed instrument, used in one exploratory correlation study [4]; no standalone validated vikṛti diagnostic exists in the reviewed set. Wearable-based wellness applications: consumer platforms exist at scale (Apple Heart Study alone enrolled 419,297 participants [18]) with strong step/HR accuracy [10, 19], but they are constitution-agnostic. Ayurveda recommendation systems: Ayurvedic mobile apps and CDAC's AyuSoft decision-support exist [1, 23], but the reviewed literature contains no system that publishes explicit, evidence-labelled rules connecting dosha assessment to daily-regimen recommendations. A 2026 review of AI in Prakriti assessment confirms that standardisation, explainability and evidence integration remain open problems [22].

11.2 Comparison of Existing Systems

Existing study / system	Prakriti	Vikriti	Wearable integration	Seasonal context	Ayurveda recommendations	Explainability
AyuSoft (C-DAC) [1]	Yes - best validated (85 items, IRR, n = 3,416)	Not in reviewed sources	No	Not in reviewed sources	Decision-support orientation; rule transparency not reviewed	Limited
Tripathi et al. 2011 [2]	Yes - 62-item instrument + scoring	No	Lab physiology only (not wearables)	Controlled (season avoided as confounder)	No	Scoring formula published
Mysore Tridosha Scale [7]	Yes - psychological domain (157 items)	No	No	No	No	Psychometric reporting
Brief Prakriti Inventory [8]	Yes - 21 items, strong psychometrics	No	No	No	No	IRT/LCA reporting
Mills et al. 2019 [4]	Brief prefix only	Yes - 60-item instrument	No	Noted as confounder only	No	Factor structure + associations published
Consumer wearable platforms (Apple/Google ecosystem) [18, 24, 25]	No	No	Yes - validated HR/steps/sleep	Weather/season passive features	No	Metric dashboards; algorithm opacity
Wearable validation literature [10, 14, 15, 17, 19]	No	No	Yes - accuracy evidence	Seasonal biology [21] separate	No	Validation metrics published
Dinacharya/CSNE knowledge base [5]	Constitution noted as context	No	No	Linked chapter (ṛtucaryā)	Yes - classical catalogue with citations	Verse-level citations
Ritucharya review [6]	Individualisation flagged preliminary	No	No	Yes - full scheme	Yes - seasonal regimen	Critical review framing
AI-in-Prakriti review [22]	Yes - meta-level	No	Not in reviewed abstract	No	Not in reviewed abstract	Called open problem
This platform (proposed)	Yes - 62-item engine	Yes - separate 60-item engine	Yes - Health Connect / HealthKit context layer	Yes - season + ṛtu-sandhi module	Yes - evidence-tagged rule engine	Every rule emits its sources + evidence level

Table 11.1 Capability comparison across existing studies/systems (based on reviewed literature; a blank cell = not present in the reviewed sources).

11.3 Limitations That Remain and What Has Not Been Studied

- No gold standard for Prakriti; physician agreement 0.20-0.40 [1].
- No psychometrically validated vikṛti instrument; only exploratory associations [4].
- No peer-reviewed evidence linking any wearable variable to dosha state; dosha-cardiovascular studies largely null [2].
- Ritucharya's direct clinical efficacy untested; constitution-based seasonal personalisation preliminary [6].
- Wearable accuracy varies by skin tone, BMI, activity type [10, 15]; vendor algorithms opaque and mutable.
- No reviewed system emits machine-readable, per-recommendation evidence levels; reproducibility of Ayurveda app logic is unstudied.

11.4 Realistic Novelty Contributions for This Platform

Proposed novelty	Supporting gap	What the platform does differently	Technical implementation	Evaluation	Novel?
Evidence-tagged rule engine	No reviewed Ayurveda app exposes per-rule evidence levels [22, 23].	Every recommendation record carries CT/DR/AI/DP tag + citations; UI shows basis beside each rule.	JSON rule schema with evidence_level + source fields; FastAPI rule evaluator (Ch. 12, 14).	Audit: 100% of rendered recommendations carry a resolvable citation; expert review of tag accuracy.	Genuinely novel as published engineering practice; combination of known parts.
Strict Prakriti/Vikriti dual-engine UX	Assessments exist separately but no reviewed app implements both with explicit separation [1, 4].	Two instruments, two engines, no merged score; UI labels "baseline" vs "current".	Two assessment modules + two DB collections + profile model (Ch. 12).	Instrument-level retest studies; user-comprehension testing.	Combination; separation itself is scientifically required, packaging is the contribution.
Honest wearable-context layer	No dosha-wearable evidence exists [2]; vendor metrics often re-presented as health facts.	Wearables shown as neutral context beside self-reports; estimates labelled; no dosha mapping anywhere.	Health Connect/HealthKit ingest + provenance flags + coverage dashboard (Ch. 8, 12).	Data-coverage analytics; accuracy-disclaimer comprehension test.	Ethical design novelty; technically combination.
Reproducible JSON rule packs	Reporting-checklist logic [3] not applied to software rules; no open rule schema found in review.	Instrument pack + rule pack published as versioned JSON with thresholds and sources.	JSON Schema files + Pytest fixtures validating every rule against schema (Ch. 12-14).	Schema-validation tests; third-party reproduction of rule outputs.	Novel application of reporting standards to app logic.
Evaluation protocol against validated comparators	Field lacks instrument benchmarks in app contexts [1, 8].	Platform pilots its 62-item engine against BPI/AyuSoft-style instruments with published statistics.	Pilot study design in Ch. 15; covariate-adjusted analysis mirroring [4].	Agreement statistics (ICC/kappa) vs comparators.	Methodological contribution; requires ethics + recruitment to realise.

Table 11.2 Proposed novelty contributions with honest novelty assessment.

Priority framing: the project must not claim to be "the first Ayurveda app" or "the first wearable-Ayurveda integration" - the literature review here is not broad enough to prove such firsts, and existing apps almost certainly exist [23]. The defensible claim is the engineering-honesty one: a fully evidence-labelled, reproducible implementation in which Prakriti, Vikriti, wearables and season are integrated without ever overstating what research supports.

12. Proposed System Architecture

12.1 Component Architecture

The architecture below matches the seven planned modules. Solid arrows show the runtime data flow: the two assessment engines write profiles; the wearable ingestion and seasonal modules write context; the recommendation engine evaluates rules over all three inputs and serves the dashboard. The knowledge database feeds rules as versioned data, so rule changes never require code changes.

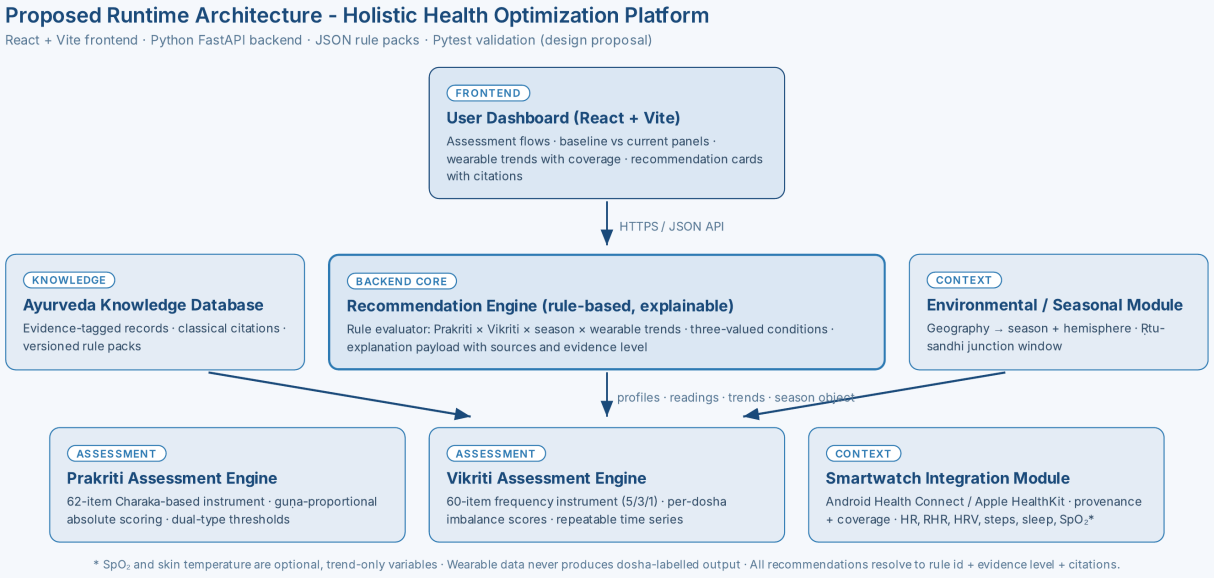


Figure 12.1 Proposed runtime architecture: assessment engines, context modules, knowledge base, rule engine and dashboard over the planned FastAPI + React stack. (Design proposal of this report.)

12.2 Module Responsibilities

Module (proposed)	Responsibility	Key data consumed / produced
Prakriti assessment engine	Serve 62-item instrument; compute guṇa scores; apply dvandvaja thresholds; persist baseline profile with instrument version.	Consumes instrument pack JSON; produces prakriti_profile {v_pct, p_pct, k_pct, type, thresholds_used}.
Vikriti assessment engine	Serve 60-item frequency instrument; compute per-dosha imbalance sums; store time series; expose trend endpoints.	Consumes vikriti instrument pack; produces vikriti_reading[] {v, p, k, context_flags, taken_at}.
Smartwatch integration module	Health Connect / HealthKit ingest; normalise units/timezones; provenance + coverage tracking; trend computation (7-day rolling, DP).	Consumes platform records; produces wearable_trend {hr, rhr, hrv, steps, active_min, sleep, stages?, spo2?, temp_delta?} with per-variable coverage.
Environmental/seasonal module	Resolve user geography → season + hemisphere; compute ṛtu-sandhi junction window; expose season object.	Produces season_context {kala, ritu, sandhi_window, hemisphere}.
Ayurveda knowledge database	Versioned store of recommendation records (Ch. 10.1) and rule packs (Ch. 14.2); evidence tags mandatory.	Produces recommendation_record[], rule_pack[].
Recommendation engine	Evaluate rules: Prakriti × Vikriti × season × wearable-trend conditions; rank by applicability; emit explanation payload with citations; handle missing inputs (Ch. 14.4).	Consumes all of the above; produces recommendation[] {rule_id, title, i18n, evidence, explanation}.
User dashboard	React surfaces: assessments, baseline vs current panels, wearable trends with coverage, recommendations with sources, education content.	Consumes API; produces nothing server-side.

Table 12.1 Module responsibilities (proposed module names; no repository filenames exist yet).

12.3 Data Flow and Trust Boundaries

Three trust rules structure the flow. First, write-once baseline: Prakriti results are versioned, never silently overwritten; a retest creates a new profile version with agreement statistics against the old. Second, one-way wearable flow: wearable data may gate or soften Ayurvedic recommendations (e.g., suppress intensity increases during sleep debt) but may never generate dosha-labelled output. Third, citation completeness: the recommendation engine must be able to explain any emitted card with rule id + evidence level + classical source + research reference; a rule without resolvable sources must fail the build (Pytest), not the user.

12.4 Proposed Data Schemas (JSON)

The following schema sketches (abbreviated) fit the planned JSON-rules approach. They are proposals to be evolved, not final designs; Pytest fixtures should validate every shipped pack against the JSON Schema at CI time.

```
{
  "instrument_pack": {
    "id": "prakriti_62_charaka_v1",
    "language": "en",
    "anchored_classical_source": "Caraka-samhita, Vimana-sthana 8.95",
    "scoring_model": {
      "type": "guna_proportional_absolute",
      "per_guna_points": 120,
      "dosha_max": {"vata": 960, "pitta": 600, "kapha": 1440}
    },
    "classification_thresholds": {
      "dual_if_secondary_ge_pct_of_primary": 50,
      "dual_if_tertiary_le_secondary_minus_pct": 25,
      "balanced_policy": "rare_fallback_rule_branch"
    },
    "items": [{"id": "Q01", "guna": "cala", "dosha": "vata",
               "text": "...", "lifespan_framing": true, "weight": 30}]
  }
}
```

```
{
  "rule_pack": {
    "id": "dinacharya_rules_v1",
    "rules": [{
      "rule_id": "DIN-ORAL-001",
      "category": "dinacharya",
      "sanskrit": "dantadhavana",
      "conditions": {"prakriti": {"vata": "sweet_or_astringent_taste_content"},
                     "vikriti": null, "season": null, "wearable": null},
      "recommendation_ref": "rec_dantadhavana_v1",
      "evidence_level": "CT",
      "sources": {"classical": "A.Hr.Su. 2/2-3; Cha.Sa.Su. 5/71-73",
                  "research": ["5"]},
      "precautions": ["gum injury caution", "not a dental treatment"]
    }]
  }
}
```

13. Research-to-Application Implementation Mapping

This chapter maps each adopted finding onto the planned stack (React + Vite frontend; Python + FastAPI backend; JSON-based rule files with storage design to be evaluated; Pytest testing). Proposed module paths use a conventional layout (frontend/src/..., backend/app/...) as suggestions only - the team should rename to match the actual repository. Testing entries name the property to test, not the assertion values, which must come from the team's own calibration.

Research finding	Source	Module / proposed path	Required data	Implementation logic	Testing required
Guṇa-proportional absolute scoring (960/600/1440)	[2]	backend/app/engine/prakriti/scoring.py	instrument_pack.json (items, weights, maxima)	sum earned points per dosha; pct = earned/max×100; store per-item audit trail.	Unit: known-answer vectors from the source questionnaire; property: pct ∈ [0,100]; formula regression vs [2].
Dvandvaja thresholds (50% / 25%)	[2]	backend/app/engine/prakriti/classify.py	thresholds from instrument pack	primary = argmax; dual if secondary ≥ 0.5×primary and tertiary ≤ secondary×0.75; order-collapse labels.	Unit tests over boundary cases (49.9%, 50.0%, 74.9%, 75.0%); golden-type fixtures.

Research finding	Source	Module / proposed path	Required data	Implementation logic	Testing required
Lifespan-stable item framing	[1, 3]	frontend/src/features/prakriti/Instrument.tsx	items with lifespan_framing flag	render every item with "over your adult life" instruction; block submit while any item answered in "current" mode.	UI test: instruction present; analytics: retest consistency (ICC target ≥ 0.83, cf. [8]).
62-item criterion-validity disclosure	[1, 2]	frontend/src/features/prakriti/ResultPanel.tsx	citation constants	display validation summary with r values and caveats on results screen.	Content test: citations render; reviewer sign-off item.
60-item vikṛti instrument (5/3/1)	[4]	backend/app/engine/vikriti/	vikriti_instrument.json (items, scale)	sum per dosha; persist reading series; compute change vs user baseline; no cutoffs.	Unit: scale math; schema validation of pack; series trend tests.
Vikriti-Psych associations as education copy	[4]	frontend/src/features/vikriti/education.ts	association table + citations	map dosha-imbalance → study-described associations, phrased as research findings; no diagnosis language.	Copy review gate; banned-phrase lint (diagnose/treat/risk).
Health Connect record ingest	[24]	backend/app/integrations/health_connect/	granted permissions; record streams	request READ_HEART_RATE, READ_RESTING_HEART_RATE, READ_HEART_RATE_VARIABILITY, READ_SLEEP, READ_STEPS, READ_OXYGEN_SATURATION as needed; normalise to UTC; provenance flag per sample.	Integration: mocked record fixtures; unit: unit/timezone conversion; coverage dashboard counts.
HealthKit ingest (iOS)	[25]	ios bridge / backend mirror of HC module	authorisation statuses per type	per-type read/share authorisation; handle permission revocation; source-device attribution.	Integration mocks; permission-change handling tests.
Estimate-vs-measured provenance + uncertainty chips	[10, 14, 15, 19]	frontend/src/components/metric/MetricCard.tsx	provenance flag from ingest	render estimate label + uncertainty chip for vendor-estimated series; hide SpO_2 absolute value (show trend only, per [17]).	Component tests; content lint for banned certainty words.
Season + ṛtu-sandhi computation	[5, 6]	backend/app/engine/seasonal/	geolocation; hemisphere table; season calendar	compute ritu + kala; flag junction window; blend adjacent regimens in window (DP).	Unit: hemisphere/date matrix; property: one active season always; sandhi flag bounded.
Knowledge record schema (evidence-tagged)	[3, 5]	backend/app/knowledge/records.json	recommendation_record schema (Ch. 10.1)	store each recommendation with classical_source, research_reference, evidence_level; enforce at schema level.	Pytest: every record has ≥1 resolvable source; evidence enum valid; build fails otherwise.
Rule evaluation with missing-input handling	DP (Ch. 14.4)	backend/app/engine/rules/evaluator.py	rule packs; profile; context	evaluate conditions; missing input → rule "unknown", not "false"; emit applicability + explanation payload.	Table-driven unit tests incl. missing/conflicting inputs; snapshot explanation payload.
Wearable trend gating of intensity advice	[5, 14, 19]	backend/app/engine/rules/rules_vyayama.json	7-day sleep trend, RHR trend (DP windows)	if sleep debt or RHR rise beyond calibrated thresholds → suppress intensity-increase rules; cite basis in explanation.	Unit: threshold boundary fixtures; calibration note in docs.

Research finding	Source	Module / proposed path	Required data	Implementation logic	Testing required
Pytest CI validation of JSON packs	[3] (spirit)	backend/tests/	all packs	schema-validate + referential integrity (rule→recommendation →source) on every build.	CI green gate; mutation smoke test.

Table 13.1 Implementation mapping (proposed; DP labels mark design proposals without a direct research source).

Build order within the backend should follow data dependencies: instrument packs → Prakriti engine → knowledge records → rule evaluator with Prakriti-only conditions → Vikriti engine → seasonal module → wearable ingest → combined rules. The frontend mirrors this: assessment flows → result panels with citations → dashboard trends → recommendation cards with sources.

14. Sample Recommendation Rules

14.1 Engine Design Principles

The engine is deterministic, rule-based and explainable - not a black-box scorer. Each rule carries explicit input conditions over four inputs (Prakriti profile, latest Vikriti reading, season context, wearable trends); an applicability decision (applicable / not_applicable / unknown-with-reason); and an explanation payload assembled from the rule's own fields. The platform deliberately uses no numerical weighting of inputs (nothing like "50% questionnaire + 30% smartwatch + 20% environment"): no reviewed research justifies any such weights, and inventing them would violate the project's evidence discipline. Rules are sequenced by a small, documented precedence (safety precautions > explicit contraindication > vikriti-correction > prakriti-maintenance > general education), itself a design proposal to be reviewed with an Ayurvedic advisor.

14.2 Rule Record Format

Each rule emits, per the project brief: input conditions; Prakriti/Vikriti inputs; wearable trend inputs; seasonal context; Ayurveda source; recommendation with Sanskrit name; English explanation; reason; precautions; evidence level. The JSON shape was shown in 12.4. The twelve worked examples below use only conditions supported by the reviewed sources; design-proposal parameters are marked DP. Evidence levels follow Chapter 2.2.

14.3 Twelve Worked Example Rules

Rule	Input conditions	Recommendation (Sanskrit - English)	Reason + classical/research source	Precautions + evidence
R1	Prakriti Vata-dominant or Vata-dual.	dantadhavana - tooth-stick taste selection: sweet/astringent content education (madhuka, khadira).	Classical taste-by-constitution table; A.Hr.Su. 2/2-3, Cha.Sa.Su. 5/71-73 via [5].	Not a dental treatment; gum injury caution. CT.
R2	Prakriti Pitta-dominant or Pitta-dual.	dantadhavana - bitter taste content (nimba).	Same classical table via [5].	Same precautions. CT.
R3	Prakriti Kapha-dominant or Kapha-dual.	dantadhavana - pungent taste content (karanja, arka) + udvartana education.	Tooth table via [5]; udvartana A.Hr.Su. 2/16 via [5].	Skin sensitivity caution. CT.
R4	Vikriti Kapha imbalance ≥ user baseline + calibrated threshold (DP); season = cool half-year.	vyāyāma dose framing: encourage consistent moderate activity within self-rated half-strength.	Classical dose rule A.Hr.Su. 2/11-14 via [5]; steps/active-min accuracy basis [19].	No max-HR prescriptions; halve advice if sleep debt. CT + DR mix.

Rule	Input conditions	Recommendation (Sanskrit - English)	Reason + classical/research source	Precautions + evidence
R5	Vikriti Vata imbalance above baseline (DP threshold); or 7-day sleep duration below user median (DP).	Suppress intensity-increase advice; offer abhyaṅga-before-bath education + earlier-bedtime regularity prompt.	Excess-exercise contraindication list A.Hr.Su. 2/14 via [5]; abhyaṅga sleep association A.Hr.Su. 2/8 via [5]; sleep-accuracy limits [14].	Wearable input is context, never a dosha readout. CT + DP.
R6	Season junction window (ṛtu-sandhi) active.	Regimen-transition education: taper previous season emphasis, adopt next gradually; keep meal/sleep times regular.	Ṛtu-sandhi concept [6, Abstract]; regularity principle [5].	Window length DP; seasonal claims labelled plausible-untested per [6]. CT.
R7	Season = peak cold (winter half) AND no Vikriti aggravation flags.	Seasonal routine emphasis: nourishing regular meals, warm bathing below neck, exercise season-appropriate.	Seasonal scheme [6]; bath temperature rules Su.Sa.Ci. 24/61 via [5].	Avoid overheating claims; not for users with heat-sensitive conditions. CT.
R8	Season = peak heat (summer) AND Prakriti Pitta-dominant.	Cooling-routine education: avoid peak-sun exertion, cooler bath emphasis, lighter evening meals; framed as classical seasonal practice.	Seasonal scheme [6]; Pitta-heat framing consistent with classical attributes [2, 5].	No dehydration-diagnosis language. CT.
R9	Meal-logging shows irregular eating (user-logged; DP) OR user reports frequent snacking.	āhāra-vidhi education: eat on true hunger, prior meal digested; two-meal regularity framing.	Su.Sa.Ut. 34/75; Cha.Sa.Vī. 1/21-25 via [5].	Explicit non-restriction disclaimer; opt-out for disordered-eating flags. CT.
R10	Vikriti Vata imbalance above baseline (DP).	Grounding-routine education: regular meal/sleep times, warm unctuous meals framing, sensory-rest breaks.	Regularity + dinacaryā principles [5]; Vata-state associations [4] as educational context only.	No anxiety claims; no medicated enemas/oil therapies (out of scope). CT + DR-mix.
R11	7-day RHR trend above user baseline (DP) OR repeated low-HRV nights (DP).	Recovery-first framing: reduce optional intensity, prioritise sleep regularity; show context card "body under load" with sources.	HRV as stress-recovery index [12]; wearable accuracy limits [10, 19].	Never a medical alert; persistent anomalies → general practitioner suggestion. DP + DR.
R12	Sleep-stages data present.	Show stage chart with uncertainty chip; education that stages are estimates; nidrā regularity emphasis.	Stage accuracy limits [14, 15]; nidrā classical anchor via [5].	No apnoea/insomnia inference; SpO_2 excluded from sleep scoring per [17]. DR + DP.

Table 14.1 Example rules (R1-R12). Wearable thresholds marked DP require platform calibration before use.

14.4 Handling Missing and Conflicting Inputs

Missing inputs: every condition is evaluated three-valued (true / false / unknown); a rule with an unknown required condition is withheld with reason "waiting for data" surfaced in a quiet "not shown now" log, so users and evaluators can see what was suppressed and why - a transparency feature, not an afterthought. Conflicting inputs: precedence is fixed (safety > contraindication > vikriti-correction > prakriti-maintenance > education) and conflicts are recorded in the explanation payload ("shown because X outweighed Y under rule precedence"), keeping the engine auditable. Wearable-vs-self-report conflicts (e.g., good sleep score but user reports poor sleep) resolve in favour of the self-report for Vikriti logic and in favour of displaying both data streams side by side - the user's lived state outranks the sensor, and the mismatch itself is shown honestly. These policies are design proposals; Chapter 15 requires expert review before release.

15. Limitations and Validation Requirements

15.1 Evidence Limitations the Team Must Own

- Instrument limits: criterion-level validation only for the 62-item tool (Pitta correlation not significant) [1, 2]; vikṛti instrument exploratory, no standalone validation [4].
- Field-level limits: no Prakriti gold standard; physician IRR 0.20-0.40; construct ambiguity [1].
- Wearable limits: HR acceptable but activity-dependent and skin-tone/BMI-sensitive [10]; EE unreliable [10, 19]; stages weak (macro-F1 0.26-0.69) [15]; SpO_2 overestimates [17]; vendor algorithms change [13].
- Seasonal limits: plausibility strong, intervention evidence weak, individualisation preliminary [6]; hemisphere dependence [21].
- Source limits: Dhiman et al. full text inaccessible to this analysis [6]; classical citations carry edition variance; some [5] modern annotations are single studies of mixed quality.
- Platform-scope limits: educational product; no diagnostic/treatment claims anywhere; classical "half-strength" dose is traditional, not physiologically validated.

15.2 Validation Requirements Before Each Release Stage

Component	Validation requirement	Basis / target	Method
Prakriti engine	Retest stability of classification.	ICC ≥ 0.83 subscale target (BPI [8]); classification agreement kappa reported.	n ≥ 30 pilot users, retest ≥ 2 weeks; report per-dosha ICC + type agreement.
Prakriti scoring	Formula conformance to [2].	Exact replication of published model.	Known-answer unit vectors; independent re-implementation check.
Vikriti engine	Score behaviour over time + comprehension.	Exploratory benchmark: no clinical cutoffs claimed [4].	Weekly-retest cohort; comprehension interviews on result language.
Wearable ingest	Coverage + provenance correctness.	Full provenance flags; no silent gaps [24, 25].	Automated coverage dashboard; device-matrix test (≥ 3 watches).
Rule engine	Explanation completeness + missing-input safety.	Every card resolvable to rule + sources [3, spirit].	Pytest: 100% rule-schema validation; explanation snapshot tests; expert rule review.
Content claims	Banned-language compliance.	No diagnosis/treatment/dosha-from-wearable language anywhere.	Phrase linter + reviewer gate on every string change.

Table 15.1 Validation matrix (targets from reviewed literature where available; DP otherwise).

15.3 What to Tell the Mentor (Evidence and Limits Summary)

A five-line briefing: (1) our questionnaire has a published lineage and a published scoring model, with correlation-level validation we disclose verbatim [1, 2]; (2) our Vikriti module follows the only peer-reviewed instrument, presented as exploratory [4]; (3) our wearable layer is built on platform-supported record types and validation literature, with estimate-labels and coverage honesty [10, 14, 15, 17, 19, 24, 25]; (4) our recommendation content is citation-first, separating classical sources from modern evidence, with seasonal claims capped at "plausible, untested" [5, 6, 21]; (5) we claim integration-with-honesty as our contribution, not scientific firsts [22, 23].

16. Prioritized Development Roadmap

16.1 Phase Plan

Phase	Build	Rationale (from this report)	Exit criteria
P0	Instrument packs + Prakriti engine + result UI with citations.	Everything downstream depends on the validated baseline instrument [1-3]; scoring model is fully specified by [2].	Formula tests green; classification boundary tests green; citation panel reviewed.
P1	Knowledge DB + rule evaluator (Prakriti-only rules) + JSON schema CI.	Citation-first architecture is the core contribution [3, 22]; earliest phase where explainability is testable.	All packs schema-valid; 100% rule source-resolvability; demo of R1-R3 + education rules.
P2	Vikriti engine + dual-panel UX + reassessment flows.	Second engine per [4]; separation is a scientific requirement [1].	Instrument math tests; trend series; banned-language gate.
P3	Health Connect ingest + coverage dashboard + context cards (no gating yet).	Platform-supported variables with known accuracy [10, 14, 19, 24]; show-only first keeps risk low.	≥ 3-device matrix shows correct provenance/coverage; timezone tests pass.
P4	Seasonal module + season rules (R6-R8).	Season is the next-lowest-risk input; honesty labels defined by [6].	Hemisphere/sandhi unit matrix green; content review.
P5	Wearable-gated rules (R4, R5, R11, R12) after threshold calibration.	Gating depends on personal baselines (DP) and calibrated thresholds; built last so data exists to calibrate.	Threshold calibration documented; suppression-log demo; expert sign-off.
P6	Validation pilot + HealthKit port + skin-temp/resp-rate explorations.	Comparator-based evaluation per Ch. 15; iOS parity [25]; exploratory variables last [17, 24].	Pilot statistics reported; iOS parity; exploratory flags labelled.

Table 16.1 Prioritised roadmap (phases, not dates; sequenced by data dependencies and risk).

16.2 Deliberate Non-Goals for v1

Defer and do not promise: ML-based Prakriti classification (unvalidated field per [1]); any dosha inference from wearable data (no evidence [2]); pulse-device integrations (no validated devices [1]); SpO_2-driven features beyond a trend line (overestimation [17]); therapeutic or herbal recommendation content (out of scope); and any alerting model beyond quiet education (notification ethics per [18] require a care pathway the platform does not have).

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